

UNIVERSITY OF SCIENCE AND TECHNOLOGY OF HANOI



WORKSPACE BOOKING SYSTEM

GROUP PROJECT REPORT

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ABSTRACT

As organizations increasingly adopt flexible and hybrid work models, the efficient management of office resources has become a critical operational priority. "Hot-desking," while offering significant benefits in real estate optimization and organizational agility, presents substantial challenges in workspace coordination and occupancy verification. This thesis presents the design and implementation of the Workspace Booking System, a comprehensive digital platform tailored for modern office management.

The system is built upon a decoupled three-tier architecture, leveraging a Next.js and React-based frontend, a modular NestJS backend, and an integrated Supabase data layer. Key technical contributions include the development of interactive, dynamic SVG-based floor maps for spatial resource discovery, a robust booking engine utilizing PostgreSQL exclusion constraints for conflict prevention, and a QR code-based check-in mechanism to bridge the gap between digital reservations and physical occupancy. Furthermore, the integration of Supabase Realtime ensures near-instantaneous synchronization across all client interfaces.

Evaluation results demonstrate that the system exceeds performance targets, with an average API response latency of 142ms and real-time state propagation of less than 0.5 seconds. The Workspace Booking System provides a scalable, secure, and user-centric solution that empowers organizations to maximize their space utilization while enhancing the daily work experience for employees in a flexible workplace environment.

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LIST OF ABBREVIATIONS

Abbreviation	Definition
API	Application Programming Interface
BaaS	Backend as a Service
CRUD	Create, Read, Update, Delete
DOM	Document Object Model
DTO	Data Transfer Object
ERD	Entity-Relationship Diagram
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
JSON	JavaScript Object Notation
JWT	JSON Web Token
MVP	Minimum Viable Product
RBAC	Role-Based Access Control
REST	Representational State Transfer
RLS	Row-Level Security
SQL	Structured Query Language
SSG	Static Site Generation
SSR	Server-Side Rendering
SVG	Scalable Vector Graphics
UI	User Interface
URL	Uniform Resource Locator
UUID	Universally Unique Identifier
WS	WebSocket

Chapter 1

INTRODUCTION

1.1 Background

In the era of digital transformation and the rapid evolution of workplace dynamics, traditional office models are undergoing a fundamental shift. The conventional approach of assigning a permanent desk to every employee is increasingly being replaced by more flexible and efficient strategies. This transition is driven by the rise of hybrid work arrangements, the need for cost optimization, and the desire to foster a more collaborative and agile work environment.

One of the most prominent strategies emerging from this shift is "hot-desking"—a workspace sharing model where employees do not have assigned desks but instead reserve or use available spaces as needed. For organizations, hot-desking offers a pathway to significantly reduce real estate overhead by maximizing the utilization of existing office space. For employees, it provides the flexibility to choose workspaces that best suit their daily tasks and encourages cross-departmental interaction.

However, the successful implementation of a flexible workspace model requires more than just a change in policy; it necessitates a robust technological infrastructure. Without a centralized and intuitive management system, the benefits of hot-desking can quickly be overshadowed by operational friction, such as uncertainty regarding desk availability, scheduling conflicts, and the inability to track actual space usage effectively.

1.2 Problem Statement

While the conceptual advantages of hot-desking are clear, organizations face several critical challenges in its practical application that stem from the lack of real-time visibility and physical verification. A primary issue is the difficulty in real-time availability tracking; employees often expend significant time and effort searching for an available desk upon arrival because there is no reliable, centralized mechanism to display which spaces are occupied or vacant across various floors. This uncertainty is further complicated by the risk of booking conflicts and overlapping reservations. Without a synchronized booking engine, multiple users may unknowingly attempt to reserve the same workspace for the same duration, leading to organizational friction and a direct loss of productivity.

Furthermore, many organizations struggle with the inefficient verification of actual attendance, often resulting in "ghost bookings" where spaces are reserved on paper but left physically empty. This lack of accountability deprives other employees of the opportunity to use those resources and skews utilization data. From a facilities management perspective, the complexity

of manually managing the layout of buildings and floors is highly labor-intensive and prone to human error. This problem is exacerbated by a lack of interactive visual guidance in standard booking tools, which typically rely on abstract lists that fail to provide the spatial context necessary for employees to make informed decisions based on their proximity to teammates or office amenities.

1.3 Significance of Study

The development of the "Workspace Booking System" is highly significant as it provides a comprehensive, technology-driven platform that directly mitigates the operational bottlenecks of modern office management. By automating the reservation process and leveraging high-performance tools like Supabase Realtime, the system enhances overall operational efficiency, eliminating the need for manual coordination and significantly reducing administrative overhead. This digital transformation allows organizations to achieve optimized space utilization; by collecting and analyzing granular usage data, facilities managers can make informed, data-driven decisions regarding office layouts and future resource allocation based on actual demand patterns.

Beyond administrative benefits, the project significantly improves the user experience through the integration of interactive SVG-based floor maps. This visual approach allows employees to locate and secure their preferred workspaces within a visually intuitive interface, reducing the cognitive load associated with traditional booking systems. The implementation of a QR code-based check-in system adds a critical layer of accountability, ensuring that bookings are physically validated upon arrival and effectively preventing the inefficiency of unoccupied reservations. Finally, by utilizing a modern and scalable tech stack—comprising Next.js, NestJS, and Supabase—the project demonstrates a sustainable architectural model capable of supporting the evolving needs of a growing enterprise in the hybrid work era.

1.4 Report Structure

This report documents the end-to-end development of the Workspace Booking System and is organized as follows:

- **Chapter 2 (Objectives):** Outlines the specific functional and non-functional goals the project aims to achieve.
- **Chapter 3 (Requirements):** Analyzes the needs of different user roles and details the system's requirements.
- **Chapter 4 (Methodology):** Describes the technical design, including the system architecture, database schema (ERD), and the implementation of key features like SVG mapping and QR verification.
- **Chapter 5 (Results and Discussion):** Presents the results of system testing and evaluates the performance of the implemented solution.
- **Chapter 6 (Conclusion and Future Work):** Summarizes the project's contributions and suggests enhancements for future iterations of the system.

Chapter 2

OBJECTIVES

The core purpose of this project is to provide a technological solution that facilitates the transition from traditional assigned-desk offices to modern, flexible working environments. This chapter outlines the primary goals of the research, the specific technical and functional objectives, the scope of the implementation, and the criteria used to measure the success of the system.

2.1 Research Objectives

The research aims to bridge the gap between flexible workspace policies and their practical, day-to-day management through the development of an integrated digital platform that harmonizes administrative control with user autonomy.

2.1.1 Main Objective

The primary objective of this thesis is to design, develop, and evaluate a robust, real-time Workspace Booking System that optimizes office space utilization and enhances the user experience in a hot-desking environment. The project seeks to create a seamless digital-to-physical workflow that ensures every square meter of office space is used effectively while providing employees with the tools they need to navigate a flexible workplace.

2.1.2 Specific Objectives

To achieve the overarching goal, the project is structured around several critical engineering and functional milestones. A fundamental objective is the design of a scalable and secure full-stack architecture utilizing modern frameworks such as Next.js, NestJS, and Supabase, specifically capable of supporting multi-tenant building and floor management. Within this architecture, the implementation of a comprehensive identity and access management system is vital to ensure data integrity through Role-Based Access Control, catering to the distinct needs of regular users, space owners, and system administrators.

A key technical focus is the development of an interactive spatial visualization interface using Scalable Vector Graphics, allowing users to interact with office layouts in a way that provides immediate environmental context. To support this, the project aims to leverage WebSocket-based communication through Supabase Realtime to ensure that all workspace availability data is synchronized instantly across all clients, thereby preventing the possibility of double-booking. Furthermore, the implementation of an automated verification workflow using QR codes is

intended to bridge the gap between digital reservations and physical occupancy. Finally, the project includes the creation of a comprehensive administrative suite that empowers facilities managers to configure resources, monitor system activity, and manage the entire lifecycle of bookings with high efficiency.

2.2 Scope of Study

This research is specifically focused on the development of a Minimum Viable Product tailored for organizations transitioning to hybrid work models, with an emphasis on core functionality and robust architecture. The implementation scope encompasses the complete lifecycle of a workspace reservation, from the initial discovery phase on interactive maps to the final check-in and eventual cancellation. The system is built upon a hierarchical data model that structures office resources into Buildings, Floors, and Workspaces, all managed within a PostgreSQL database environment. To ensure accessibility, the platform is developed as a responsive web application optimized for both desktop planning and mobile on-site verification.

To maintain a clear and focused research direction, certain advanced features have been excluded from the current MVP. These include the implementation of dynamic, time-limited QR codes, which remain a target for future security hardening. Additionally, the current scope does not include integrations with external enterprise calendars or communication tools, nor does it delve into complex predictive analytics or the management of ancillary resources like meeting rooms and parking. By concentrating on these core pillars, the project ensures a stable and high-performance foundation for a modern workspace management solution.

2.3 Success Criteria

The success of the Workspace Booking System is measured through a combination of functional, performance, and usability metrics. From a functional perspective, the system must demonstrate the ability to handle the complete booking lifecycle without any instances of data inconsistency or resource loss. Performance-wise, a critical benchmark is the real-time propagation of state changes, which should reflect across all active clients with a latency of less than one second, ensuring that the visual representation of desk availability is always accurate. Furthermore, the system is evaluated on its usability, aiming for a streamlined process where a user can locate and book a workspace on an interactive map with fewer than four clicks. Finally, the accuracy of the verification mechanism is essential, as the system must correctly validate a user's physical presence at the designated workspace through the QR code scanning process.

Chapter 3

REQUIREMENTS ANALYSIS

3.1 Overall System Requirements

The Workspace Booking System is a multi-tenant platform designed to manage physical workspace assets efficiently. The system serves three primary user roles: Super Admin, Workspace Manager, and General User.

3.1.1 Web Application (User & Manager Interface)

The web application provides a responsive interface for users to interact with the workspace ecosystem. Functional requirements include:

- **Authentication & Authorization:** Secure login managed via Supabase Auth. The system enforces Role-Based Access Control (RBAC) to differentiate between guests, space owners, and administrators.
- **Workspace Discovery:** Interactive SVG floor maps allow users to visualize office layouts and check real-time availability of desks and meeting rooms.
- **Booking Lifecycle:** Users can reserve spaces for specific time slots, while the system ensures data integrity by preventing overlapping reservations.
- **QR Check-in:** Verification of physical presence at the workspace via static QR code scanning, which updates the booking status to "Checked-in".

3.1.2 Admin Dashboard (System Oversight)

The administrative side of the system provides the tools necessary for infrastructure maintenance and organizational management:

- **System Configuration (Super Admin):** Global management of the building hierarchy (Buildings, Floors) and oversight of all user accounts and roles.
- **Space Onboarding (Workspace Manager):** Ability for space owners to propose new workspaces, upload floor layouts, and manage the booking flow for their specific assets.
- **Approval Workflows:** A mechanism for Super Admins to vet "Space Owner" requests and approve/reject new workspace submissions.

3.2 Use Cases

3.2.1 General User Use Case

The General User represents the typical guest or employee who utilizes the platform to find and book workspaces. Their primary interactions focus on the booking lifecycle and personal account management.

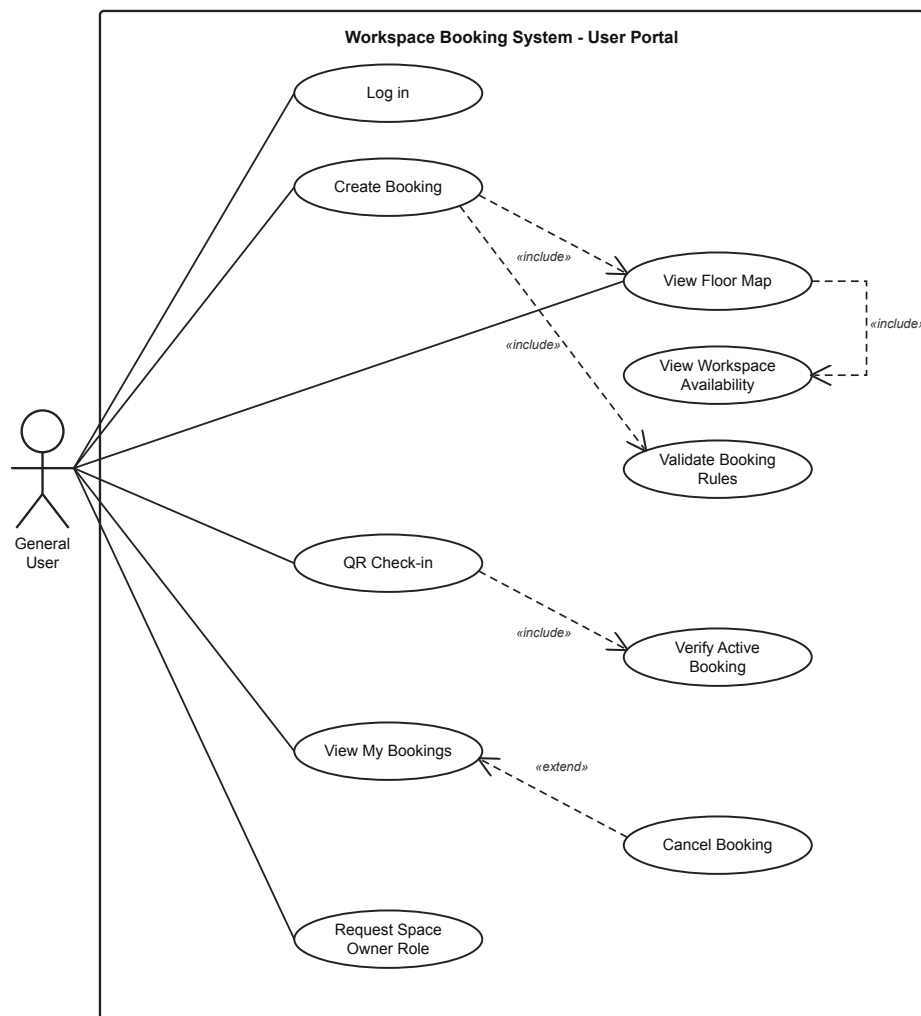


Figure 3.1: Use Case Diagram - General User

ID	Use Case Name	Description
UC-USR-01	Log in	Secure authentication to the portal.
UC-USR-02	View Floor Map	Browse buildings, floors, workspaces, and current booking state on the SVG floor map.
UC-USR-03	Create Booking	Reserve a space for a selected time slot.
UC-USR-04	Check-in	Scan QR code at the desk to verify arrival.
UC-USR-05	View My Bookings	Review personal booking history and current reservations.
UC-USR-06	Cancel Booking	Cancel an eligible personal booking.
UC-USR-07	Request Space Owner Role	Submit a request to become a Workspace Manager.

Table 3.1: Summary of General User Use Cases

3.2.2 Workspace Manager Use Case

The Workspace Manager (Space Owner) is an elevated user who can use the normal booking features and monitor reservation activity for the spaces they own. In the current implementation, workspace publication and QR asset generation are controlled through the Super Admin console.

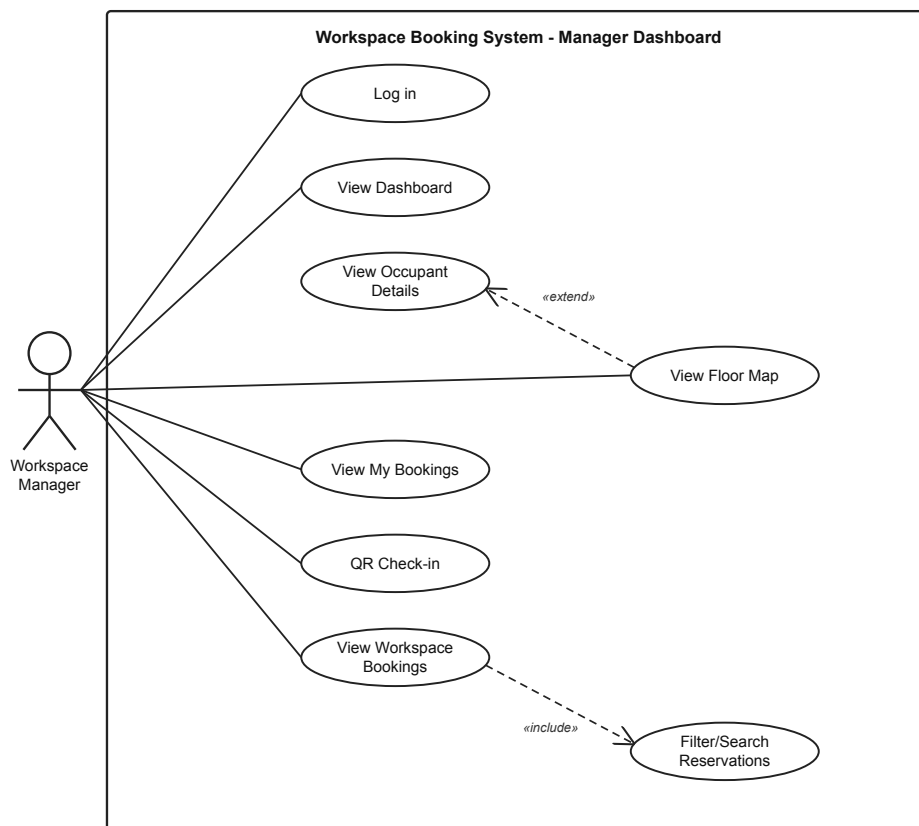


Figure 3.2: Use Case Diagram - Workspace Manager

ID	Use Case Name	Description
UC-MGR-01	View Dashboard	Access the authenticated dashboard and role-aware navigation.
UC-MGR-02	View Floor Map	Browse available workspaces and booking state like a regular user.
UC-MGR-03	View My Bookings	Review their own reservations.
UC-MGR-04	Check-in	Check in to their own reservation using the workspace QR code.
UC-MGR-05	View Workspace Bookings	Monitor reservations for workspaces they own.

Table 3.2: Summary of Workspace Manager Use Cases

3.2.3 Super Admin Use Case

The Super Admin possesses global authority over the entire system. They are responsible for the foundational building data and the vetting of users and workspace submissions to maintain system quality.

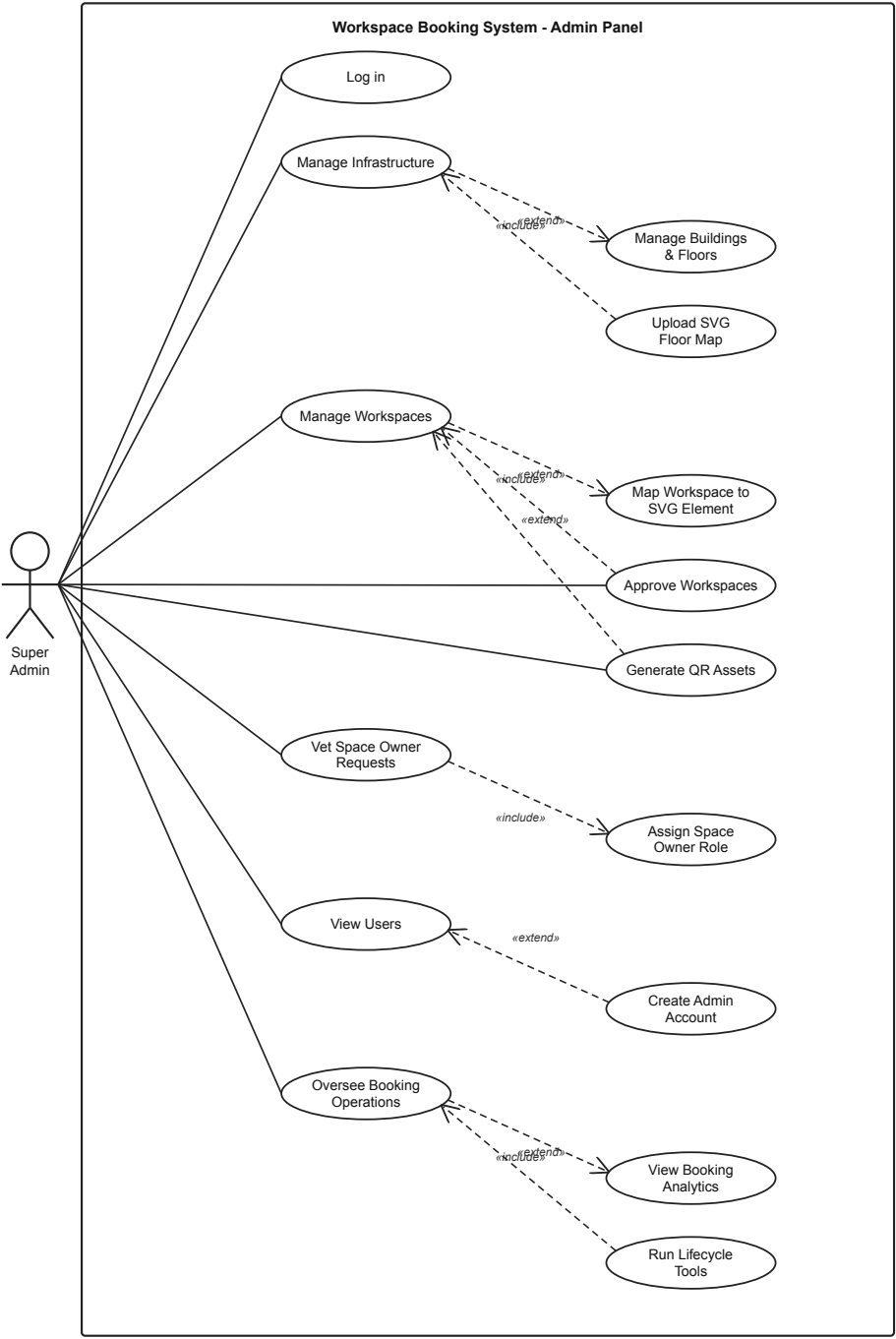


Figure 3.3: Use Case Diagram - Super Admin

ID	Use Case Name	Description
UC-ADM-01	Manage Infrastructure	Configure Buildings and Floors.
UC-ADM-02	Upload SVG Floor Map	Upload and inspect SVG floor layouts for workspace mapping.
UC-ADM-03	Manage Workspaces	Create workspace records and map them to SVG element IDs and QR values.
UC-ADM-04	Vet Space Owner Requests	Approve or reject requests for the 'space_owner' role.
UC-ADM-05	Approve Workspaces	Audit and approve/reject submitted workspace records.
UC-ADM-06	Generate QR Assets	Generate, copy, and download physical QR labels for workspaces.
UC-ADM-07	View Users / Create Admin	View registered users and create trusted admin accounts.
UC-ADM-08	View Booking Analytics	View global booking activity, status counts, and utilization indicators.
UC-ADM-09	Run Booking Lifecycle Tools	Execute no-show and completion maintenance actions.

Table 3.3: Summary of Super Admin Use Cases

3.3 Use Case Specifications

3.3.1 General Operations

Log in

Goal: To provide secure access based on user roles.

Primary Actor: User, Manager, Admin.

Main Success Scenario:

1. User enters email and password.
2. System verifies credentials via Supabase.
3. System redirects to the appropriate dashboard (User Portal or Admin Dashboard).

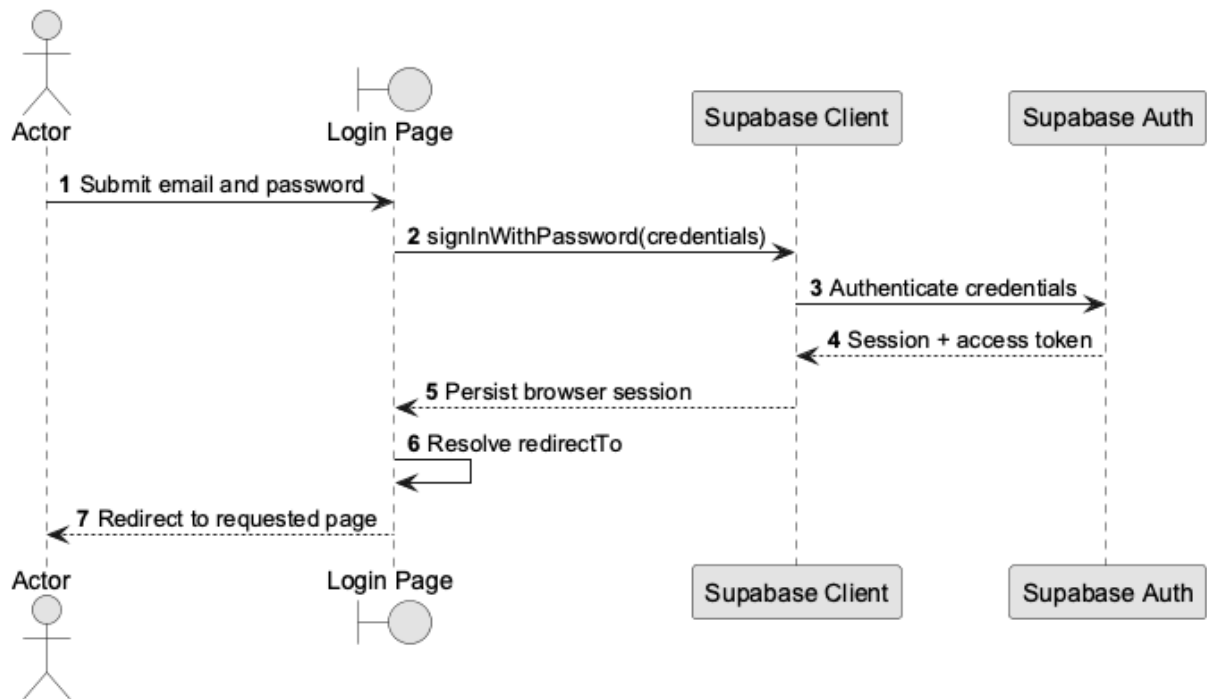


Figure 3.4: Sequence Diagram - Authentication

3.3.2 User & Booking Operations

View Floor Map

Goal: To browse available workspaces and booking state.

Primary Actor: General User, Workspace Manager, Super Admin.

Main Success Scenario:

1. Actor opens the floor map screen.
2. System loads buildings, floors, approved workspaces, and time-based booking state.
3. Actor selects a building, floor, date, and time range.
4. System displays the SVG floor map and workspace availability.

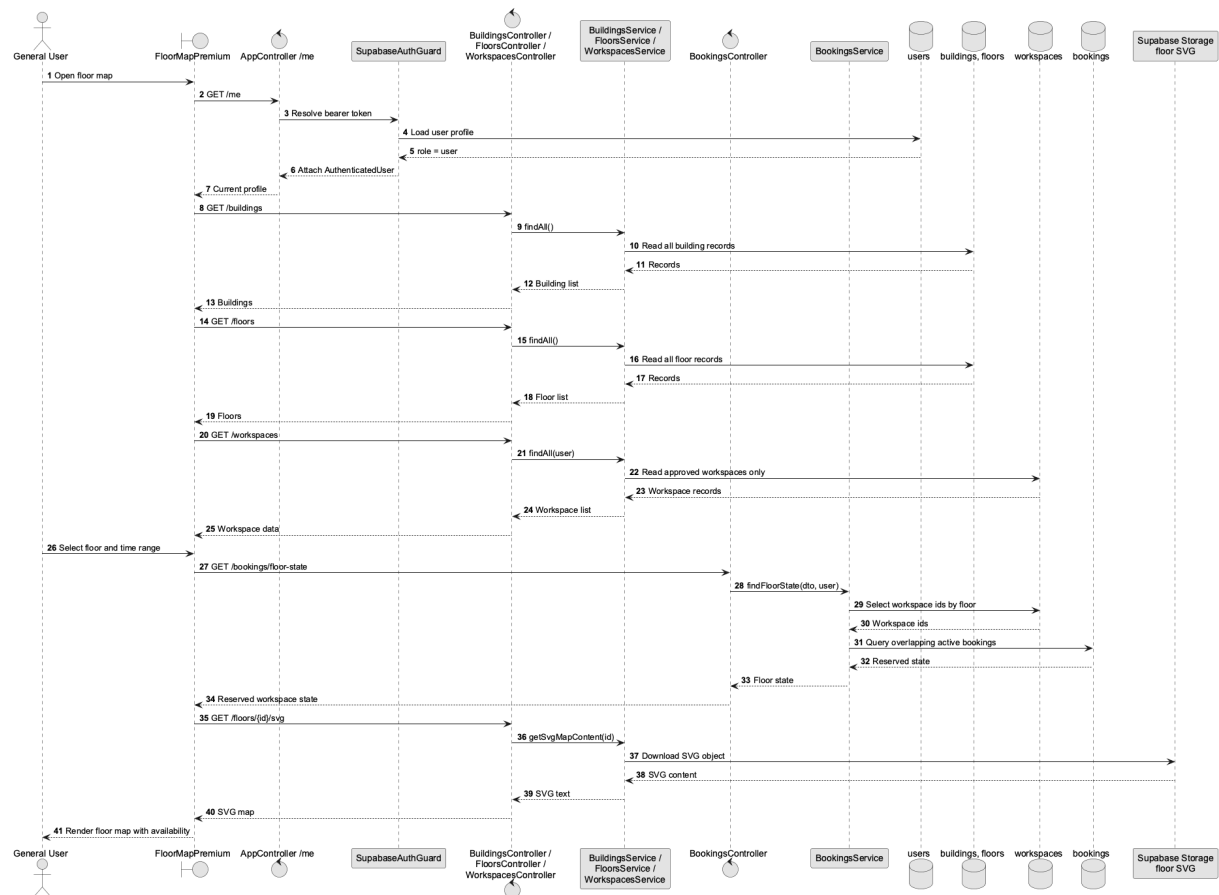


Figure 3.5: Sequence Diagram - View Floor Map

Create Booking

Goal: To reserve a workspace.

Primary Actor: General User, Workspace Manager.

Main Success Scenario:

1. Actor selects a workspace from the SVG floor map.
2. Actor specifies the booking time range.
3. System validates booking policy and database conflicts.
4. System confirms the booking.

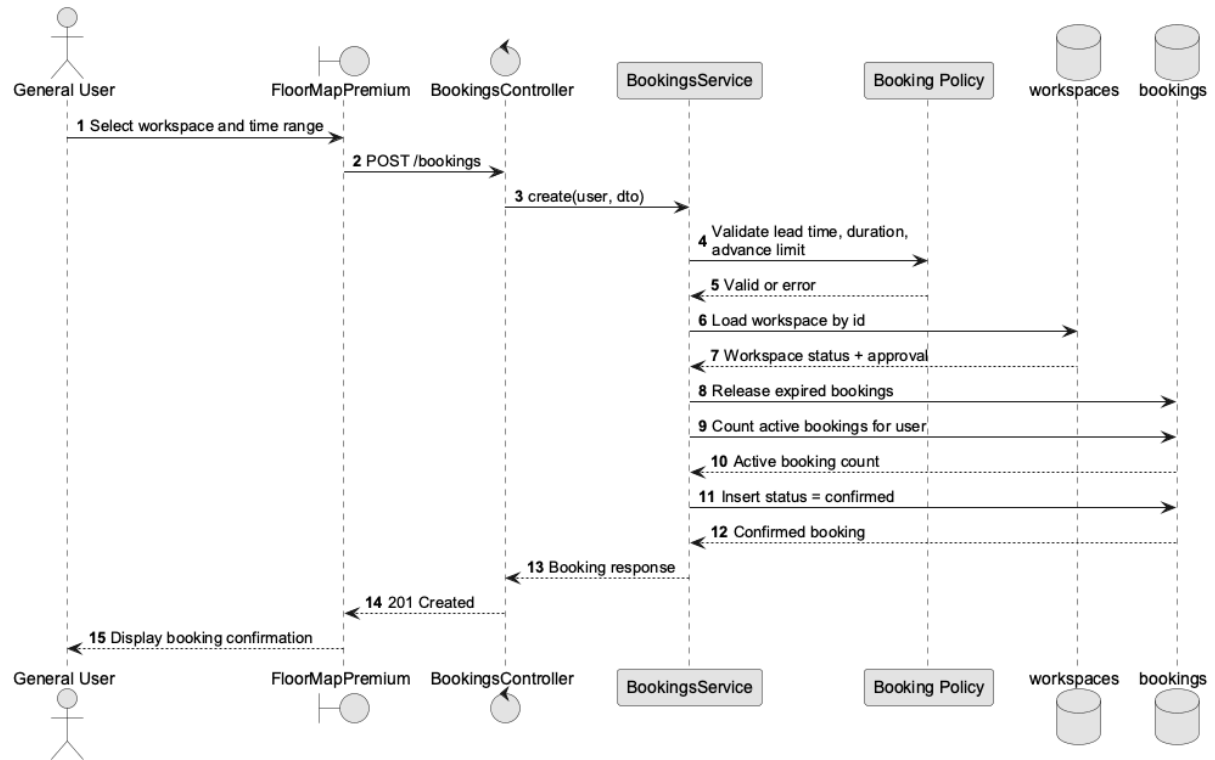


Figure 3.6: Sequence Diagram - Create Booking

View My Bookings

Goal: To review personal reservations.

Primary Actor: General User, Workspace Manager.

Main Success Scenario:

1. Actor opens the bookings screen.
2. System lists the actor's booking history and current reservations.

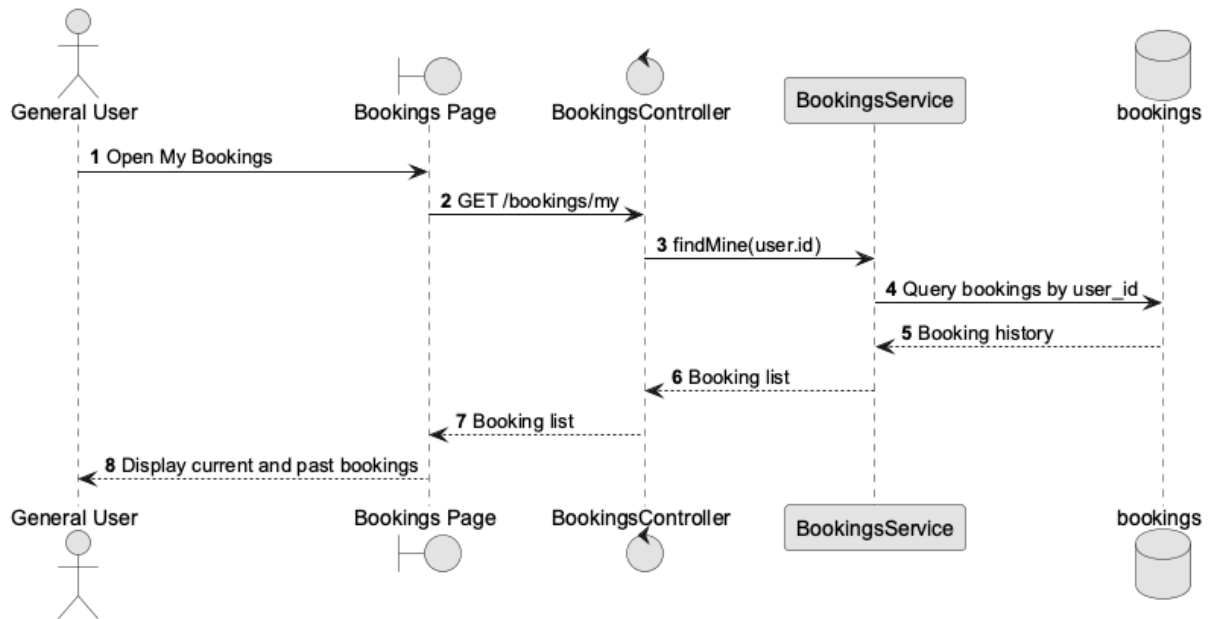


Figure 3.7: Sequence Diagram - View My Bookings

Cancel Booking

Goal: To cancel an eligible personal reservation.

Primary Actor: General User, Workspace Manager.

Main Success Scenario:

1. Actor opens the personal bookings screen.
2. Actor selects an eligible confirmed booking.
3. System validates ownership and booking status.
4. System updates the booking status to "Cancelled".

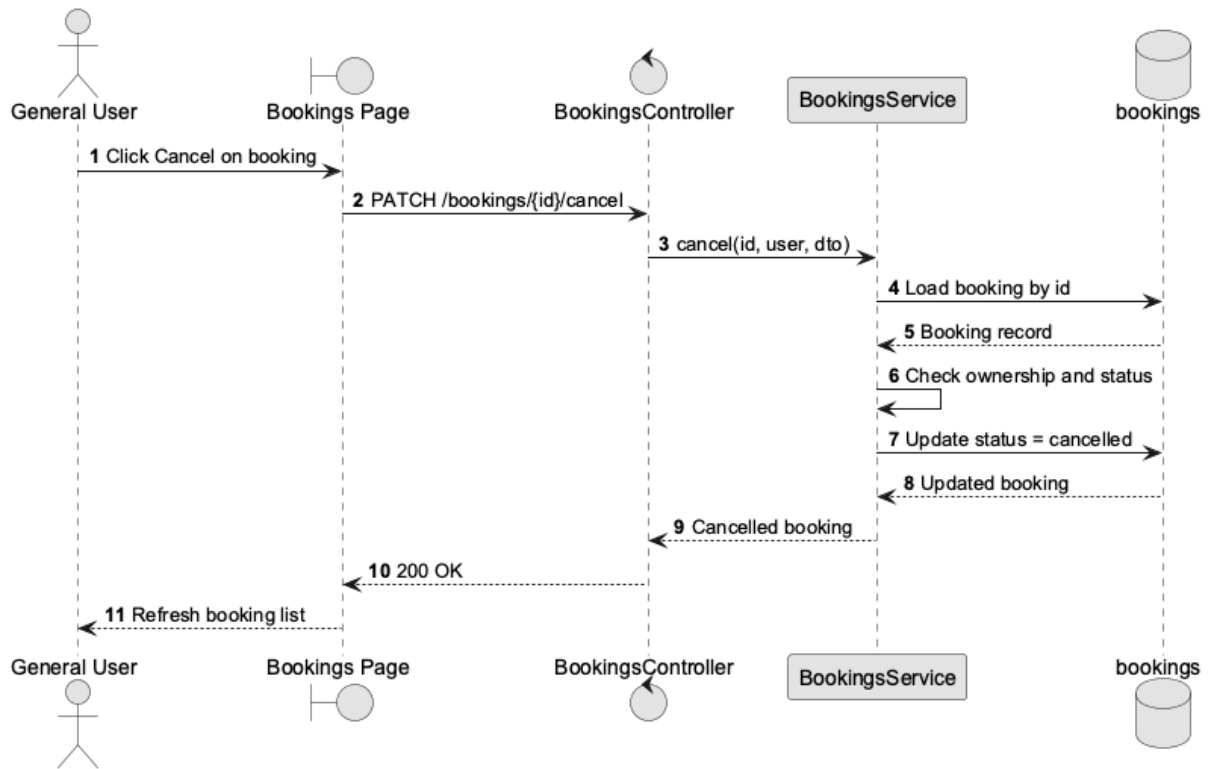


Figure 3.8: Sequence Diagram - Cancel Booking

QR Check-in

Goal: To verify arrival at a desk.

Primary Actor: General User, Workspace Manager.

Main Success Scenario:

1. Actor scans the desk QR code or enters the QR value manually.
2. System validates the workspace QR value and active booking status.
3. System updates booking to "Checked-in".

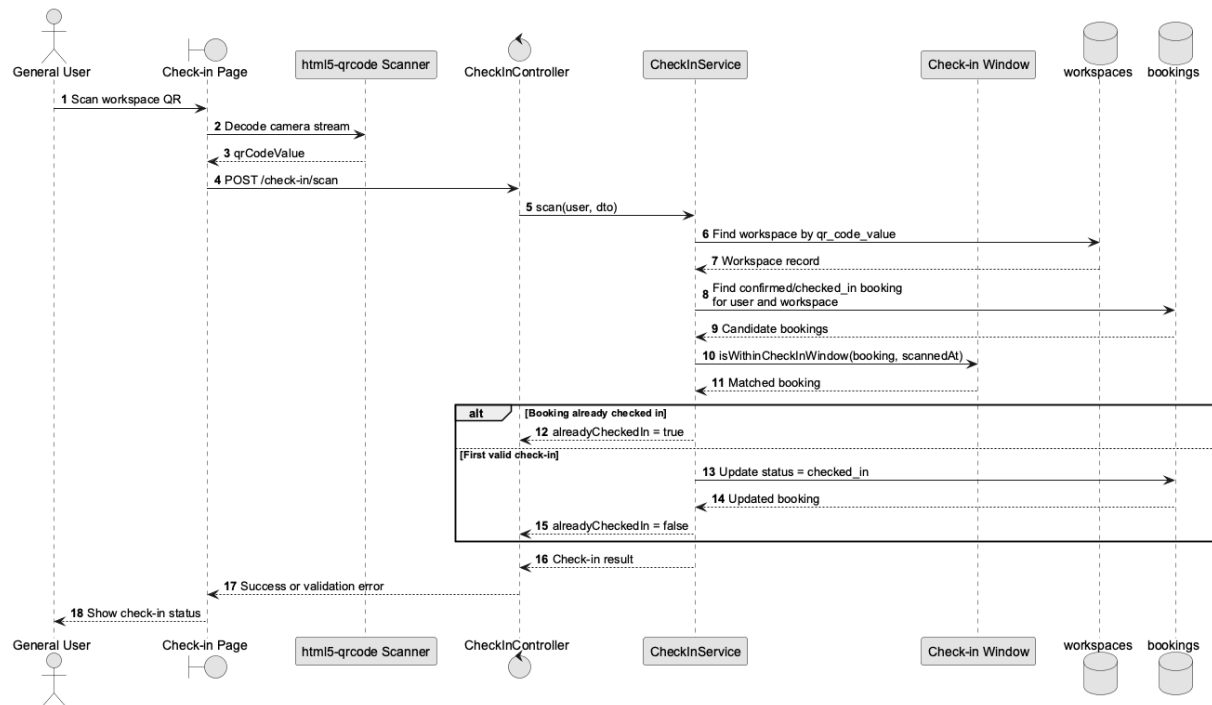


Figure 3.9: Sequence Diagram - QR Check-in

Request Space Owner Role**Goal:** To request permission to monitor owned workspace reservations.**Primary Actor:** General User.**Main Success Scenario:**

1. User opens the dashboard.
2. User submits a Space Owner access request.
3. System stores the request with "Pending" status.
4. Super Admin reviews the request and the system updates the user's role if approved.

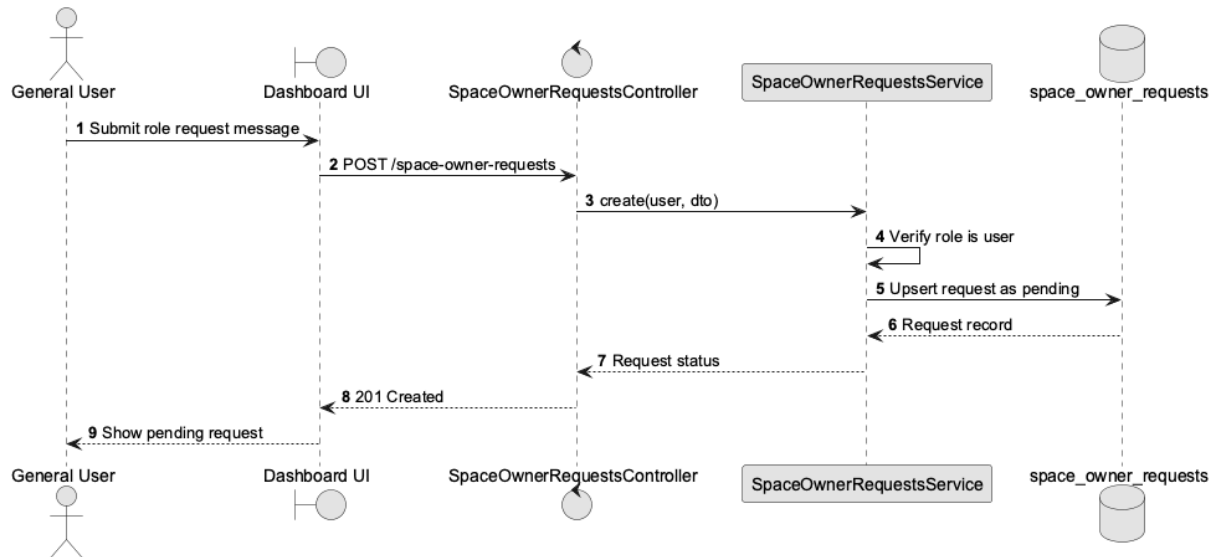


Figure 3.10: Sequence Diagram - Request Space Owner Role

3.3.3 Workspace Manager Operations

View Dashboard

Goal: To access role-aware navigation and dashboard data.

Primary Actor: Workspace Manager.

Main Success Scenario:

1. Manager opens the dashboard.
2. System resolves the manager's authenticated role.
3. System loads personal booking and request state.
4. System displays the manager dashboard.

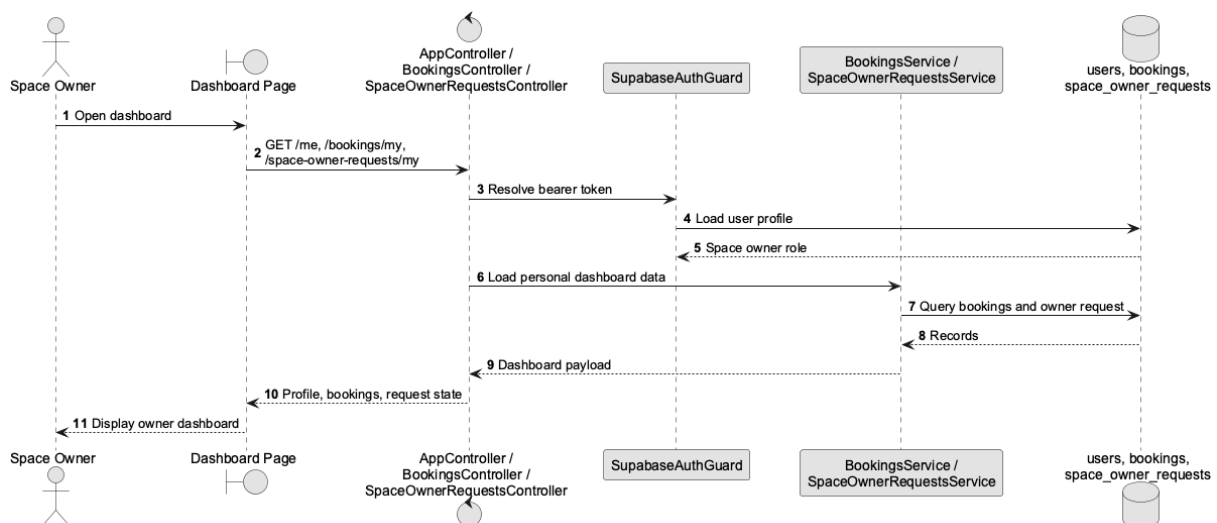


Figure 3.11: Sequence Diagram - Manager Dashboard

Manager View Floor Map

Goal: To browse workspaces and occupant details available to managers.

Primary Actor: Workspace Manager.

Main Success Scenario:

1. Manager opens the floor map screen.
2. System loads buildings, floors, approved workspaces, and owned workspace records.
3. Manager selects a floor and time range.
4. System displays the floor map and occupant details permitted for the manager role.

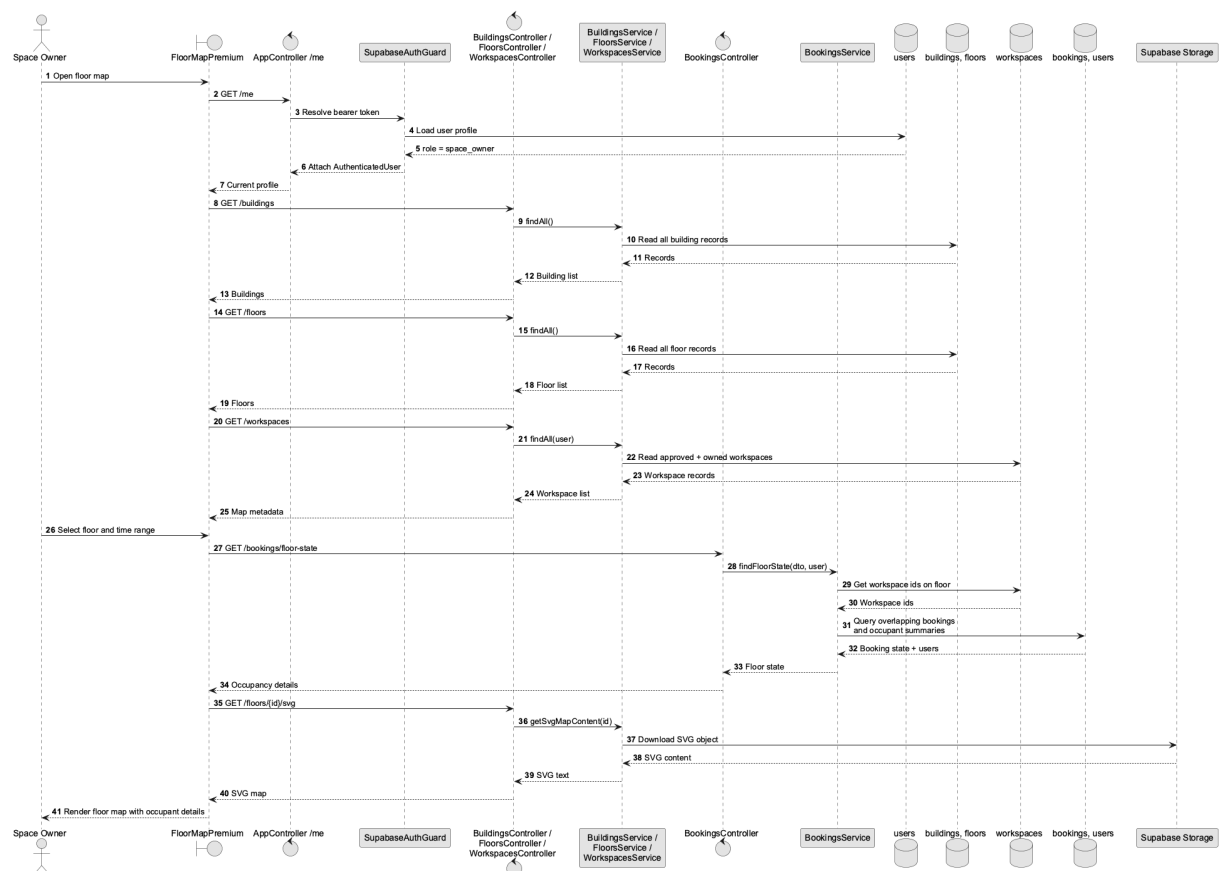


Figure 3.12: Sequence Diagram - Manager View Floor Map

Manager View My Bookings

Goal: To review the manager's own reservations.

Primary Actor: Workspace Manager.

Main Success Scenario:

1. Manager opens the bookings screen.
2. System loads bookings associated with the manager's account.
3. System displays the manager's personal reservations.

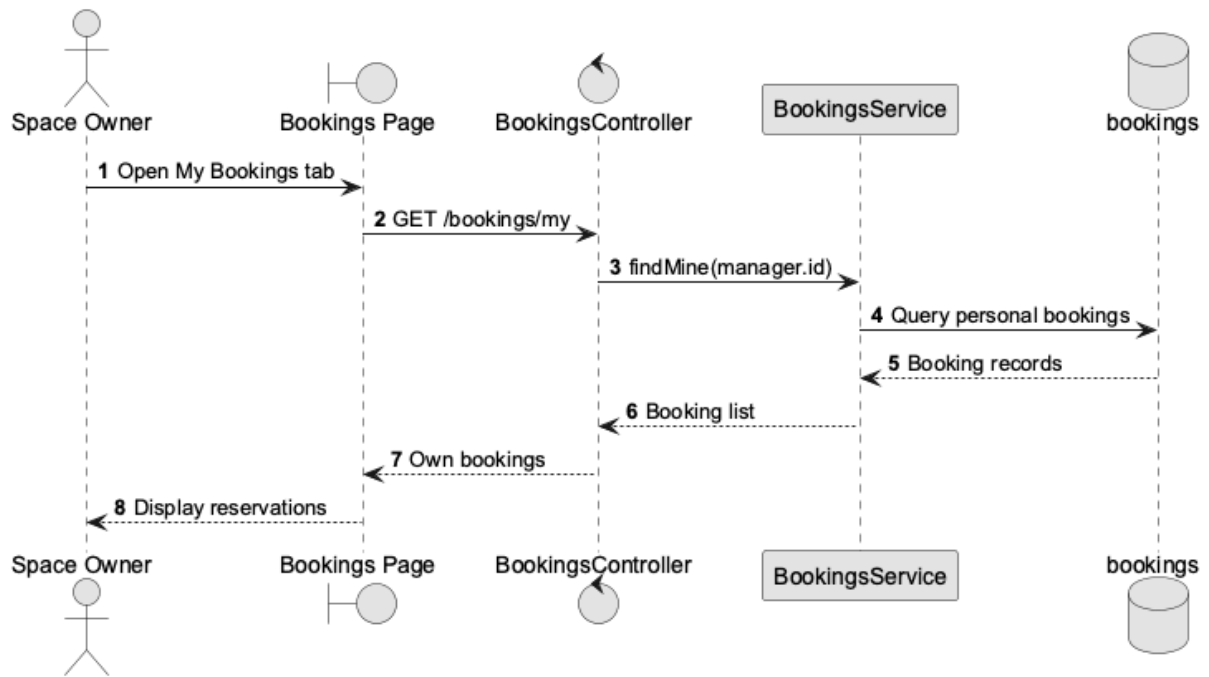


Figure 3.13: Sequence Diagram - Manager View My Bookings

Manager QR Check-in**Goal:** To check in to the manager's own reservation.**Primary Actor:** Workspace Manager.**Main Success Scenario:**

1. Manager scans the workspace QR code.
2. System validates the QR value and the manager's active booking.
3. System updates the booking to "Checked-in".

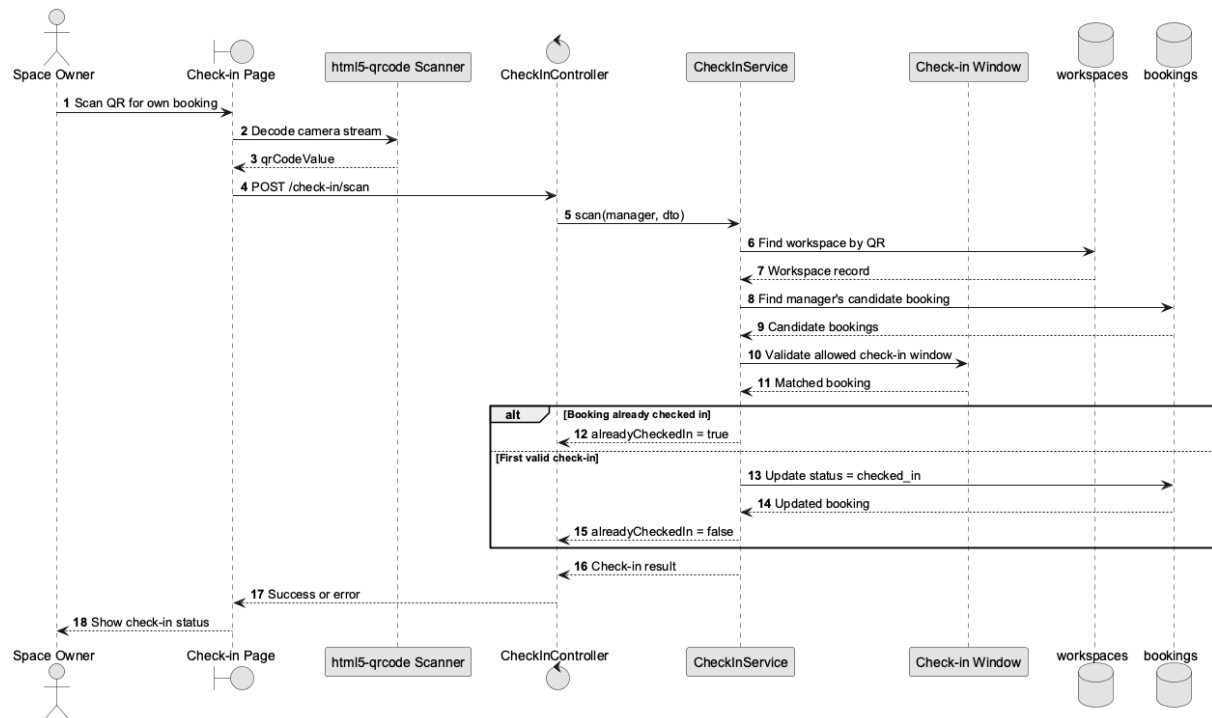


Figure 3.14: Sequence Diagram - Manager QR Check-in

View Workspace Bookings

Goal: To monitor reservations for owned workspaces.

Primary Actor: Workspace Manager.

Main Success Scenario:

1. Manager opens the bookings screen.
2. Manager switches to the managed bookings view.
3. System lists bookings associated with the manager's owned workspaces.
4. Manager searches and reviews reservation records.

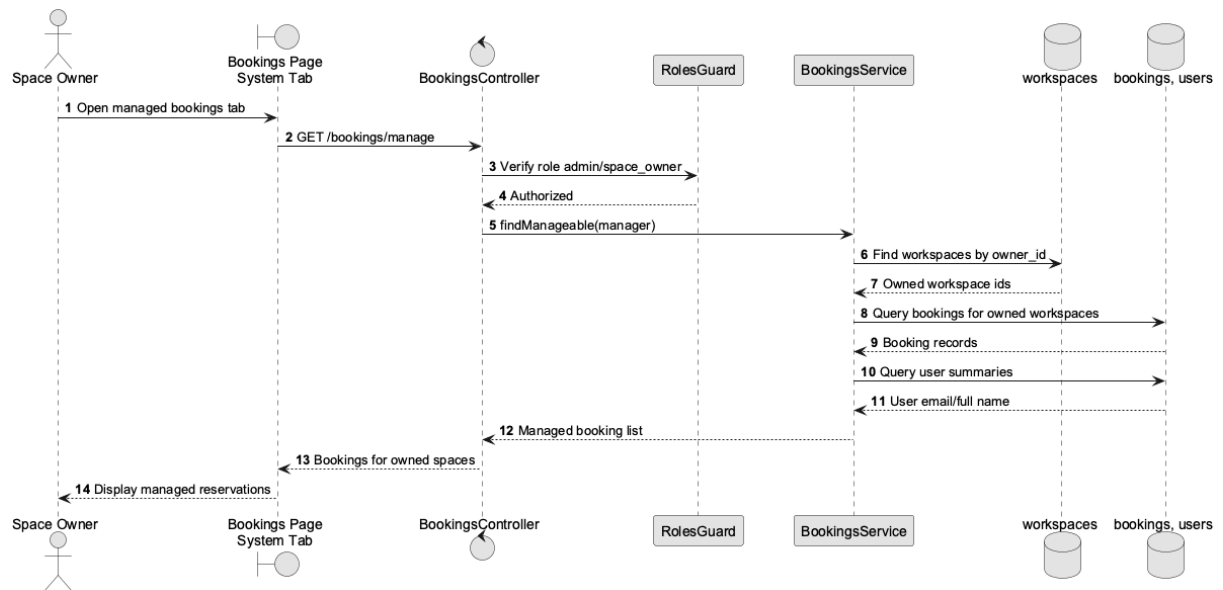


Figure 3.15: Sequence Diagram - View Bookings for My Spaces

3.3.4 Super Admin Operations

Manage Infrastructure

Goal: To configure the physical building and floor hierarchy.

Primary Actor: Super Admin.

Main Success Scenario:

1. Admin creates or reviews buildings and floors.
2. System stores the infrastructure records.
3. System displays the updated building and floor hierarchy.

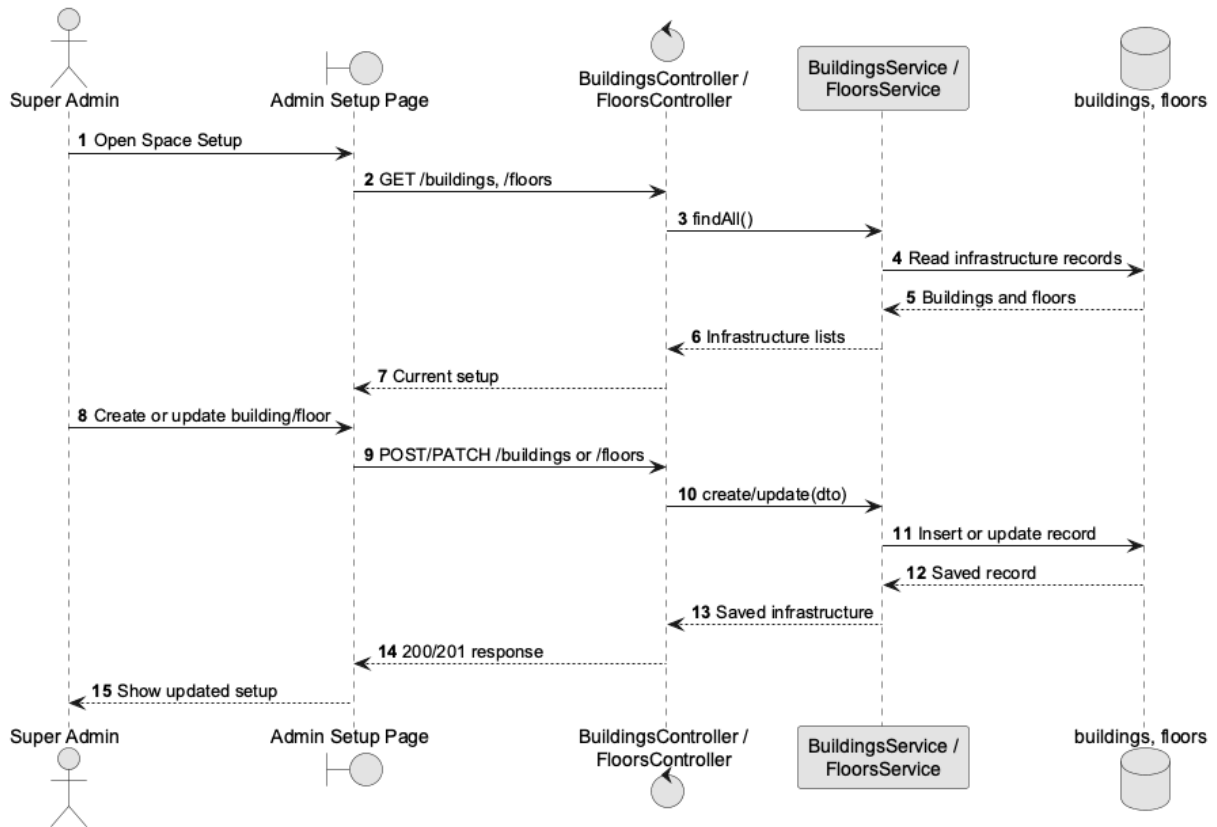


Figure 3.16: Sequence Diagram - Manage Infrastructure

Upload SVG Floor Map

Goal: To attach an SVG layout to a floor for workspace mapping.

Primary Actor: Super Admin.

Main Success Scenario:

1. Admin selects a floor and uploads an SVG file.
2. System validates the file and stores it in Supabase Storage.
3. System updates the floor's SVG map reference.
4. System exposes the SVG content for element inspection.

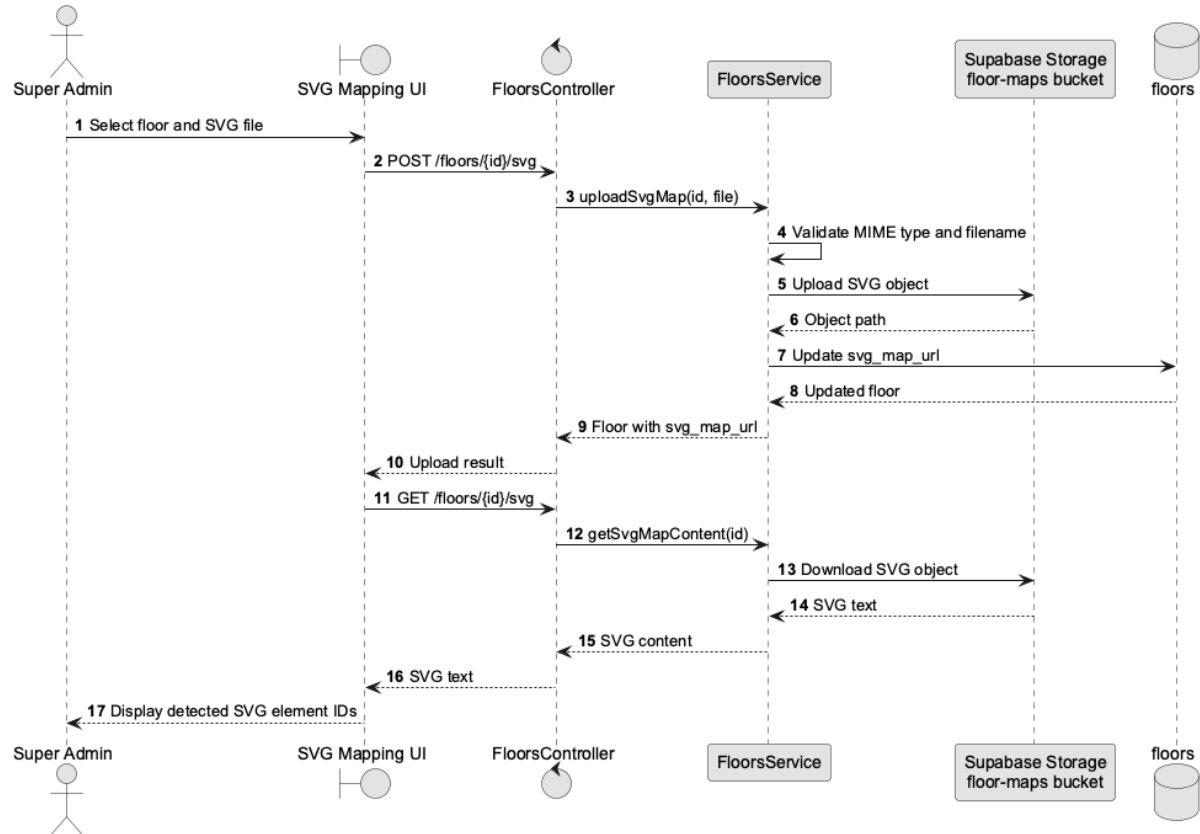


Figure 3.17: Sequence Diagram - Upload SVG Floor Map

Manage Workspaces

Goal: To create workspace records and map them to SVG elements.

Primary Actor: Super Admin.

Main Success Scenario:

1. Admin creates a workspace with floor, SVG element ID, type, capacity, and QR value.
2. System stores the workspace record.
3. System reloads and displays the updated workspace list.

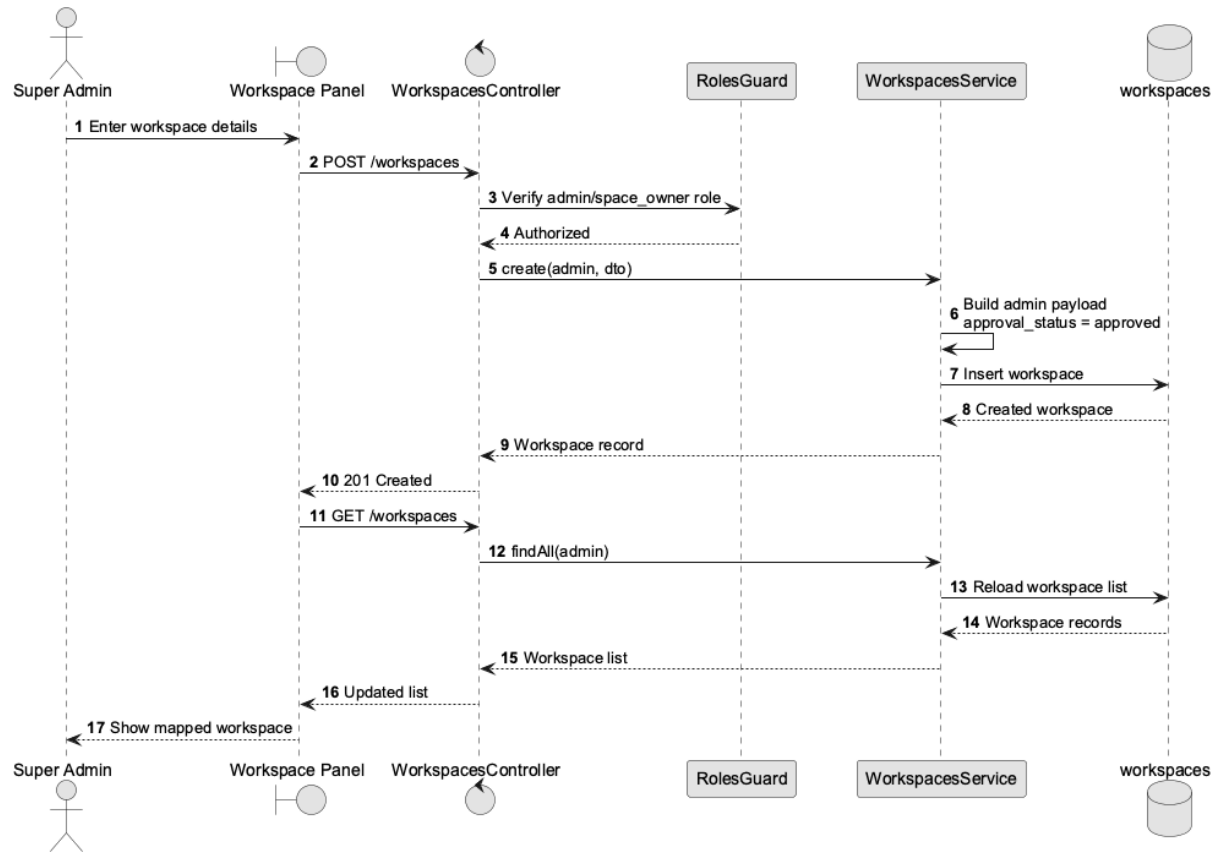


Figure 3.18: Sequence Diagram - Manage Workspaces

Approve Workspaces

Goal: To control publication status for workspace records.

Primary Actor: Super Admin.

Main Success Scenario:

1. Admin reviews pending workspace records.
2. Admin approves, rejects, or hides a workspace.
3. System updates approval metadata and publication status.

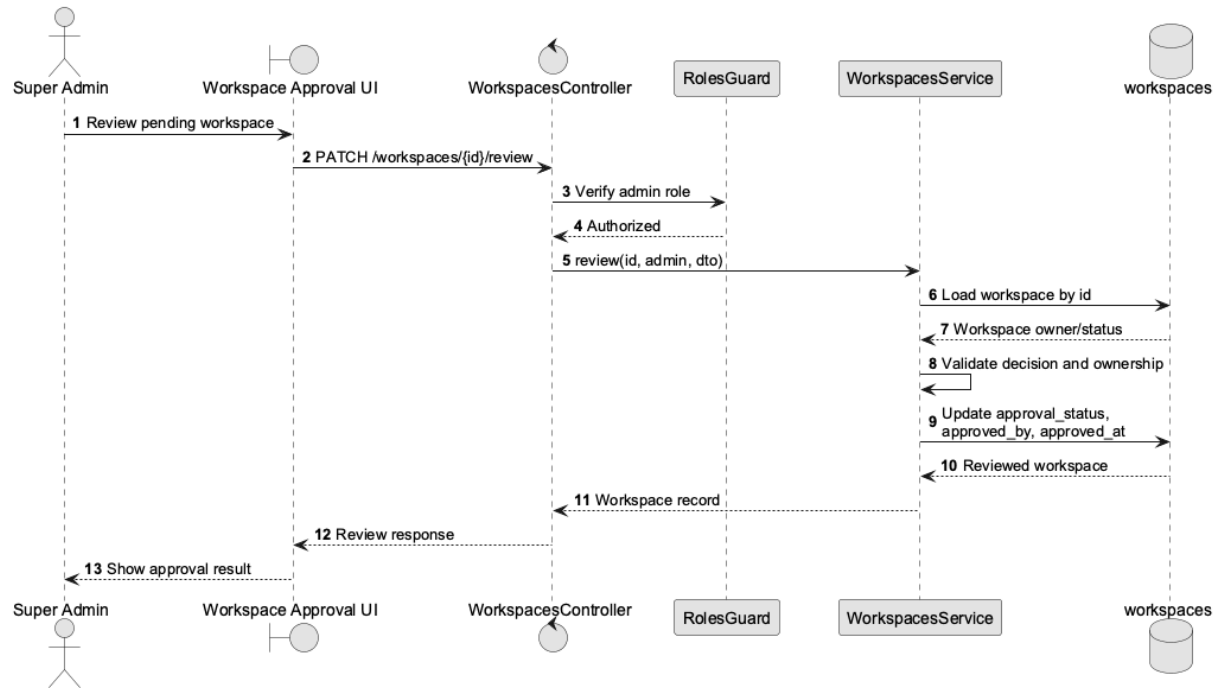


Figure 3.19: Sequence Diagram - Approve or Reject Workspace

Vet Space Owner Requests

Goal: To review requests for the Workspace Manager role.

Primary Actor: Super Admin.

Main Success Scenario:

1. Admin opens the owner request queue.
2. System lists submitted requests.
3. Admin approves or rejects a request.
4. If approved, system updates the requester role to "space_owner".

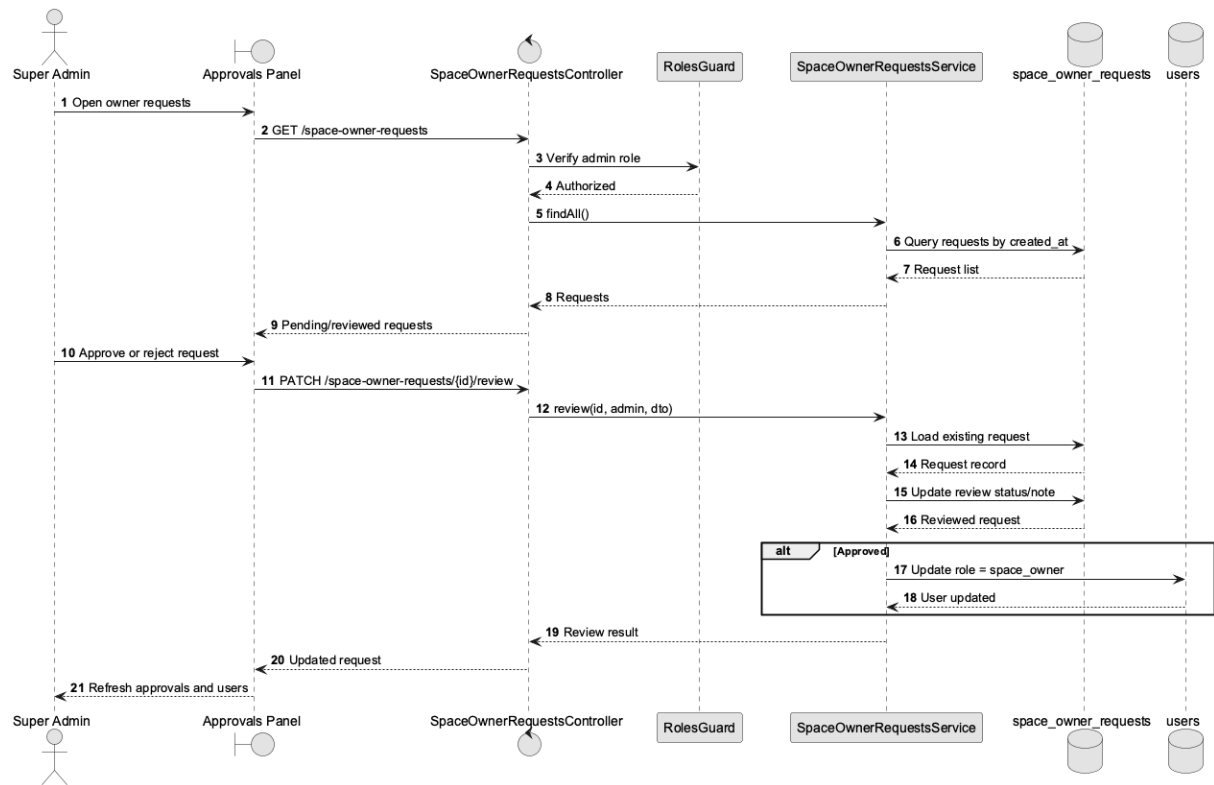


Figure 3.20: Sequence Diagram - Vet Space Owner Requests

Generate QR Assets

Goal: To prepare QR labels for physical workspace check-in.

Primary Actor: Super Admin.

Main Success Scenario:

1. Admin opens the QR assets screen.
2. System loads workspaces, floors, and buildings.
3. Admin selects a workspace and exports or copies its QR label.

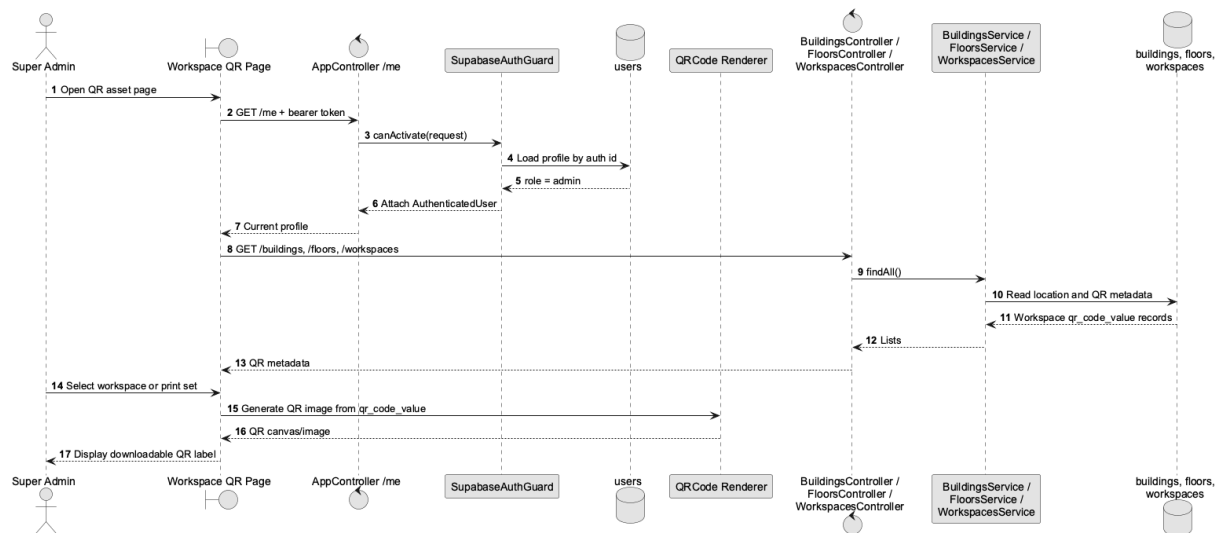


Figure 3.21: Sequence Diagram - Generate QR Assets

View Users / Create Admin

Goal: To inspect accounts and create trusted administrator users.

Primary Actor: Super Admin.

Main Success Scenario:

1. Admin opens User Management.
2. System lists users and their roles.
3. Admin provides email, temporary password, and optional full name.
4. System creates a new admin account.

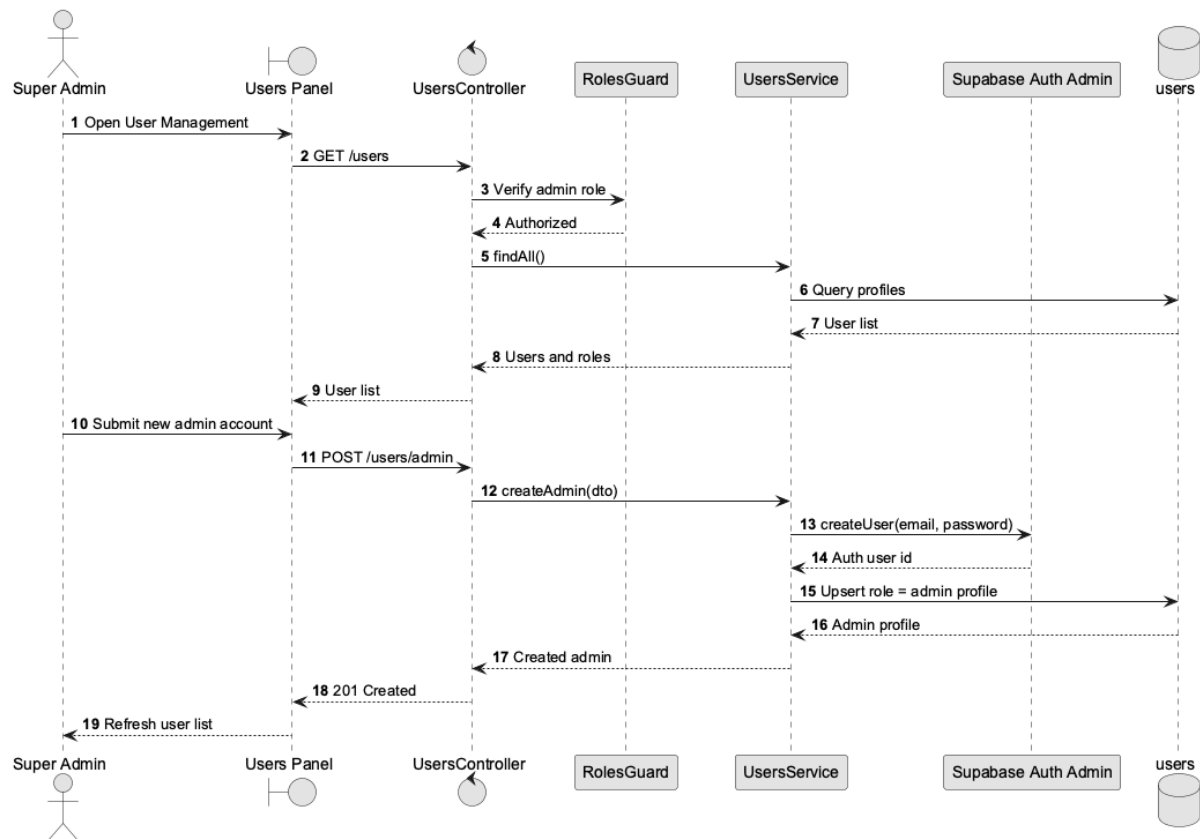


Figure 3.22: Sequence Diagram - View Users and Create Admin Account

View Booking Analytics

Goal: To monitor global booking activity and utilization indicators.

Primary Actor: Super Admin.

Main Success Scenario:

1. Admin opens the system bookings view.
2. System loads global bookings and workspace metadata.
3. System calculates status counts, utilization, and volume indicators.
4. System displays the analytics dashboard.

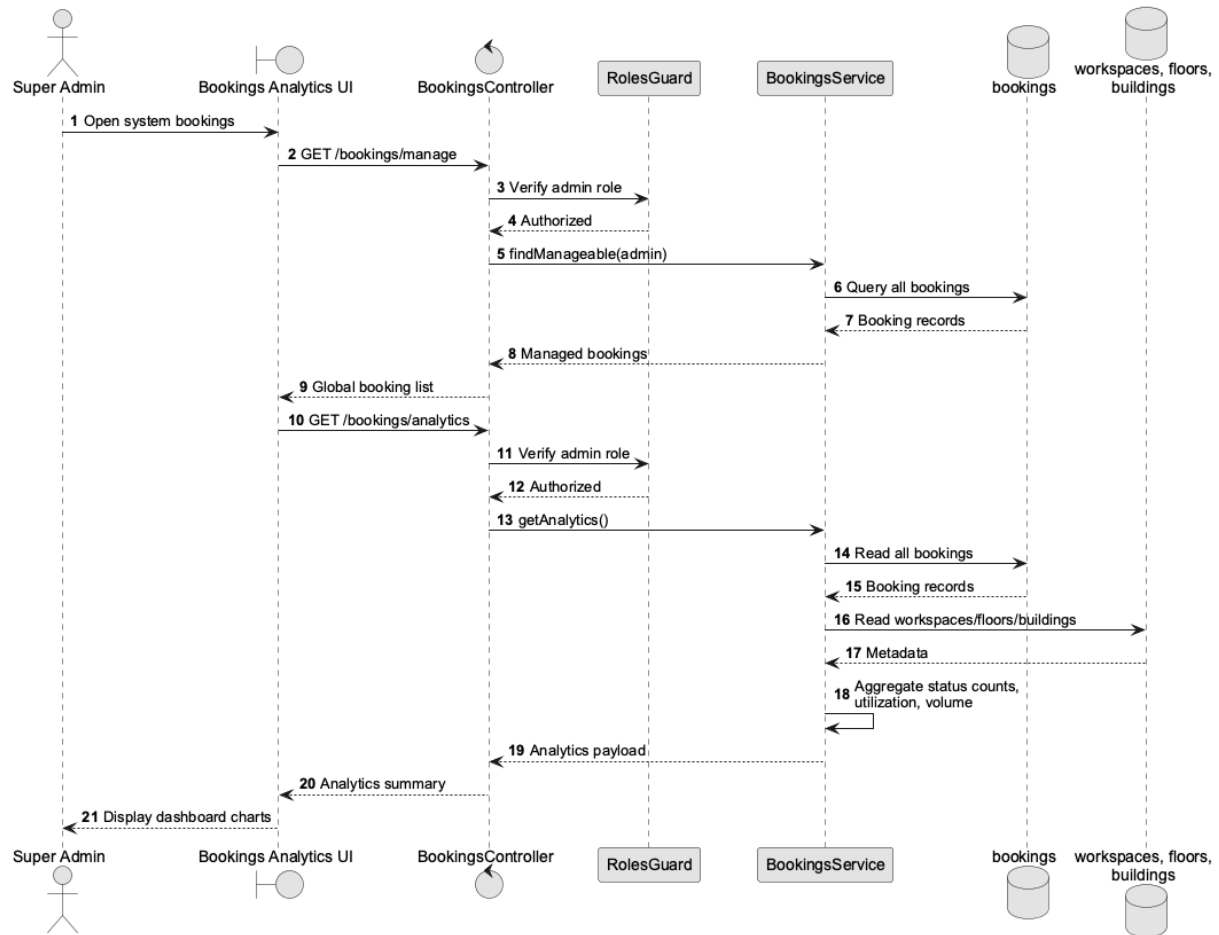


Figure 3.23: Sequence Diagram - Booking Analytics

Run Booking Lifecycle Tools

Goal: To run no-show and completion maintenance actions.

Primary Actor: Super Admin.

Main Success Scenario:

1. Admin opens lifecycle tools from the system bookings view.
2. Admin runs no-show or completion processing.
3. System identifies eligible bookings.
4. System updates booking statuses and returns the affected count.

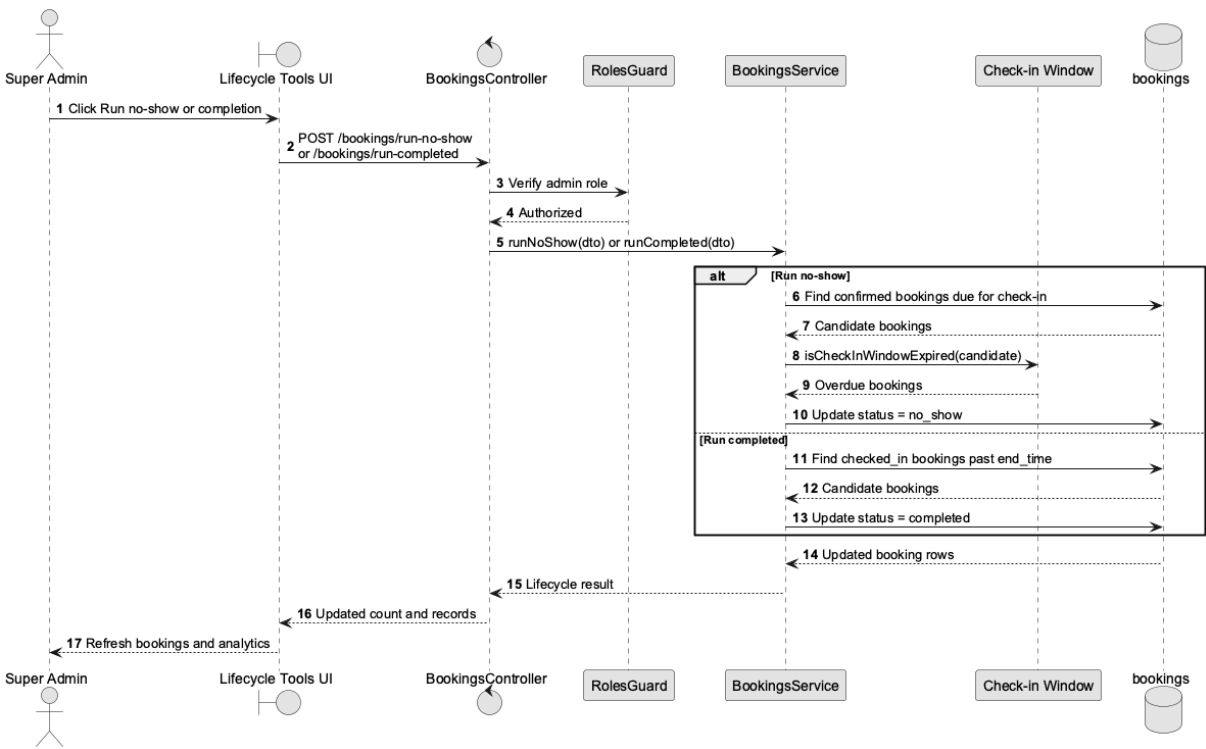


Figure 3.24: Sequence Diagram - Booking Lifecycle Tools

Chapter 4

METHODOLOGY

This chapter provides a detailed exposition of the technical framework, architectural design, and implementation strategies employed in the development of the *Workspace Booking System*. It justifies the selection of the technology stack and delves into the core modules and engineering logic that enable interactive workspace management.

4.1 Tools and Techniques

The selection of the technology stack for the *Workspace Booking System* was guided not only by functional requirements like real-time spatial synchronization and interactive floor mapping, but also by specific engineering goals: establishing a modular and scalable backend, guaranteeing database-level transactional safety, maintaining high security-by-design, and ensuring development efficiency inside a unified monorepo. This section provides a detailed justification for the selected tools, frameworks, and techniques compared to common industry alternatives.

4.1.1 Backend Services & API Layer

- **NestJS Framework (vs. Express.js):** While Express.js is widely used for building lightweight Node.js backends, it lacks standard architectural patterns, which frequently leads to disorganized and unmaintainable codebases as the system's business logic expands. NestJS, a progressive TypeScript framework, was selected because of its highly structured, modular architecture. By employing built-in design patterns like dependency injection, decorators, and middleware, NestJS enforces a strict separation of concerns. This architectural rigour is crucial for managing the complex business rules of the booking system, such as handling seat check-ins, validation policies, and roles, ensuring high testability and long-term maintainability.
- **TypeScript (vs. JavaScript):** The entire monorepo is written in TypeScript. Compared to vanilla JavaScript, TypeScript's static typing acts as a powerful build-time safeguard, catching class mismatch, syntax, and configuration errors before deployment. Furthermore, using TypeScript enables the sharing of core types, interfaces, and data transfer objects (DTOs) between the NestJS backend and the Next.js frontend through the local packages/shared workspace. This single source of truth minimizes API contract discrepancies and accelerates feature development.
- **Passport & Passport JWT (vs. Custom Session Auth):** Instead of coding an ad-hoc session store or a custom JWT verification layer—both of which are prone to subtle security

bugs like token leakage or insufficient validation—the backend implements authentication using *Passport.js* and its JWT strategy. By integrating with Supabase’s secure authentication mechanism, Passport allows the system to validate client JSON Web Tokens (JWT) seamlessly on each incoming request and securely extract the user’s identity and roles for Role-Based Access Control (RBAC) without incurring session lookup overhead.

- **NestJS Swagger (vs. Manual Documentation):** To streamline developer cooperation, API documentation is auto-generated using `@nestjs/swagger`. By using decorators directly within controllers and DTOs, NestJS generates interactive Swagger UI endpoints automatically. This ensures the API specifications are always in sync with the codebase, eliminating the overhead of manual API documents and allowing the frontend to integrate endpoints with absolute precision.

4.1.2 Database & Backend-as-a-Service (BaaS)

- **Supabase BaaS (vs. Custom Self-Hosted Infrastructure):** Rather than spending significant engineering overhead deploying, configuring, and securing isolated components (PostgreSQL, custom OAuth systems, S3 storage, and WebSocket brokers), the system relies on Supabase. As an integrated Backend-as-a-Service (BaaS) platform, Supabase bundles PostgreSQL database management, OIDC-compliant authentication, file storage, and real-time event broadcasting. This unified environment guarantees production-grade reliability out of the box, letting the development team focus entirely on domain-specific application features while reducing infrastructure complexity.
- **PostgreSQL with GIST Indexing (vs. NoSQL/MongoDB):** To manage workspace booking transactions, transactional safety is paramount. PostgreSQL was chosen over NoSQL alternatives (like MongoDB) due to its superior ACID compliance and advanced constraint mechanisms. Specifically, to prevent the critical problem of double-bookings, the system utilizes PostgreSQL exclusion constraints backed by a GIST index. This database-level safeguard guarantees that no two bookings for the same workspace can have overlapping duration intervals, enforcing schedules directly within the persistence layer.
- **Supabase Realtime (WebSockets vs. HTTP Polling):** Dynamic office management requires instant visual feedback on workspace occupancy. Supabase Realtime leverages WebSockets to broadcast database changes (INSERT, UPDATE, DELETE) on the bookings table instantly. By subscribing to these changes, the frontend updates the floor maps in real time, avoiding the heavy network overhead and delay associated with traditional short or long HTTP polling, thereby eliminating booking collisions during peak traffic.
- **PostgreSQL Row-Level Security (RLS) & Database Triggers:** Security is enforced directly inside PostgreSQL using Row-Level Security (RLS) policies. This ensures that clients communicating directly with Supabase are constrained by strict access control rules (e.g., standard users can only read bookings but cannot delete or modify administrative settings). Additionally, database-level triggers automate state updates, such as marking workspace availability, maintaining data consistency independently of the application code.
- **Supabase Storage (vs. AWS S3/MinIO):** Floor maps are stored as Scalable Vector Graphics (SVG) within Supabase Storage. This integrated solution provides secure, authenticated access control via RLS policies while exposing highly optimized presigned URLs.

This eliminates the latency and security overhead of setting up separate cloud storage wrappers like AWS S3 or self-hosted MinIO clusters.

4.1.3 Presentation Layer (Frontend Web App)

- **Next.js App Router (vs. Create React App):** The web application is built on Next.js 16.2. Its hybrid rendering architecture combines Server-Side Rendering (SSR) for fast initial loads and Static Site Generation (SSG) for high-performance static pages. Next.js's native App Router simplifies nested route layouts and directory structures, while built-in features like image optimization and API middleware protect administrative routes, resulting in a highly performant and secure portal.
- **React 19 (Component-Based UI):** Using React 19's virtual DOM allows for seamless component reconciliation and state management. React's modularity is essential for rendering floor plans, where each workspace, desk, or room is modeled as an independent, interactive component that dynamically responds to user interaction and real-time state changes.
- **Tailwind CSS v4 (vs. CSS Modules):** Styling is driven by Tailwind CSS v4's utility-first framework. It enables rapid interface layout while avoiding bloated CSS files. Utility classes enable smooth visual transitions on SVGs, allowing the floor plans to represent desk availability states (e.g., available, occupied, or currently selected) with distinct, consistent colors matching the Tailwind theme configuration.
- **Framer Motion (vs. Raw CSS Transitions):** To elevate the user interface into a premium, responsive experience, Framer Motion is utilized. It powers smooth transitions between floors, high-fidelity micro-interactions, and visual feedback when booking seats, creating an intuitive and engaging environment for users.
- **QR Code Scanning and Generation (vs. Manual Entry):** To build a secure check-in verification bridge, the web app utilizes `qrcode` for generating unique desk QR codes and `html5-qrcode` for camera-based scanning. This allows employees to check in to their workspaces instantly by scanning a physical QR code at their desk, providing proof of physical presence.

4.1.4 Infrastructure, Quality Assurance, & Testing

- **Docker & Docker Compose (vs. Manual Host Deployment):** Containerization ensures consistency across local development, staging, and production environments. Using multi-stage Dockerfiles for the Next.js web application and the NestJS backend, along with Docker Compose, allows the entire multi-tiered application—including the local Supabase environment—to be orchestrated in a single, predictable command.
- **Playwright (vs. Cypress):** End-to-end (E2E) testing is critical to verify the booking flow and real-time updates. Playwright was selected over Cypress for its native support for multiple browser contexts, superior speed, and robust handling of complex canvas and SVG elements. Playwright allows for reliable simulations of user interactions, including real-time synchronization between two active browser sessions and QR code check-in flows.

- **Jest (vs. Mocha):** Unit and integration testing for backend services are executed via Jest. Built into NestJS CLI, Jest allows developers to mock database dependencies, isolate service layers, and execute high-speed unit tests to ensure business validation policies remain robust.

4.2 System Architecture

This section details the architectural design of the *Workspace Booking System*, focusing on the structural components, communication protocols, and engineering rationale behind key decisions. The architecture is modeled using the C4 software engineering model, capturing both the high-level container organization and the low-level API modular structure.

4.2.1 Overall System Architecture (C4 Container View)

The overall system architecture is designed around three core principles: *separation of concerns*, *scalability through modularity*, and *security-by-design*. The system adopts a modern **three-tier client-server model** where the presentation layer (web frontend), business application layer (REST API), and data persistence/infrastructure layer (Supabase Backend-as-a-Service) are strictly decoupled. This organization is represented in the Container C4 Diagram in Figure 4.1.

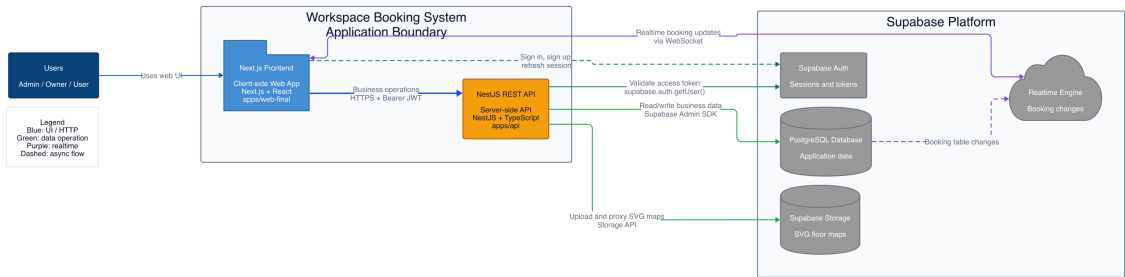


Figure 4.1: C4 Container Diagram of the Workspace Booking System

The system is decomposed into the following core containers and platform modules:

- **Next.js Frontend Container:** The primary client-side web application serving all user roles (Standard Users, Space Owners, and Administrators). Built with Next.js App Router and React, it renders interactive SVG floor maps, handles booking requests, and accesses user cameras to perform physical QR check-ins.
- **NestJS REST API Container:** The protected backend application built with NestJS and TypeScript. It acts as the central business orchestrator, enforcing validation policies, processing booking states, and executing administrative transactions. It secures its endpoints using Passport JWT guards and reads/writes persistent data using the Supabase Admin SDK.
- **Supabase Auth Service:** A managed identity provider that handles user registrations, sign-ins, and session refresh tokens. The Next.js client directly communicates with Supabase Auth to retrieve JSON Web Tokens (JWT), which are then attached as Bearer tokens in headers when calling the NestJS API.

- **PostgreSQL Database:** The relational database layer that stores structured tables for buildings, floors, workspaces, bookings, and check-in logs. To maintain transactional consistency, it implements unique constraints and exclusion indexing to completely prevent double-booking issues.
- **Supabase Realtime Engine:** A WebSocket-based publisher-subscriber service. Instead of routing floor-map occupancy queries through the NestJS API at short intervals, the Next.js client establishes a direct WebSocket subscription to the PostgreSQL bookings table through Supabase Realtime. When a desk is booked or checked in, database changes are instantly broadcast to all connected clients, enabling real-time visual updates.
- **Supabase Storage Container:** An integrated object storage bucket used specifically for managing floor-map assets. Floor maps are uploaded and stored as SVG files, which are securely retrieved by the client via optimized, authenticated public URLs.

By decoupling operations, this three-tier architecture allows each layer to scale and evolve independently. For example, the Next.js presentation layer can undergo substantial UI modifications or be migrated to a native mobile app without affecting backend business logic, provided the API endpoints contract remains unchanged.

4.2.2 Backend Application Architecture (C4 Component View)

To prevent the operational overhead and networking complexities associated with a distributed microservices setup (such as service-to-service authentication, event synchronization, network latencies, and managing multiple databases), the NestJS backend is engineered as a **Modular Monolith**. As shown in the Component C4 Diagram (Figure 4.2), the application logic is strictly divided into decoupled, domain-specific NestJS modules sharing a unified TypeScript codebase.

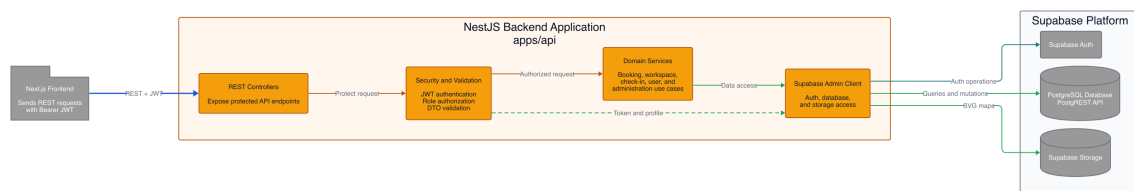


Figure 4.2: C4 Component Diagram of the NestJS REST API Backend

The internal structure of the NestJS API consists of the following key components:

- **Auth Layer (Guard System):** Implements custom NestJS guards (`SupabaseAuthGuard` and `RolesGuard`). It acts as a gateway on protected endpoints, intercepting incoming HTTP requests to validate the Bearer JWT with Supabase Auth. It then decodes the payload to extract roles (User, Space Owner, Admin) and enforces fine-grained access control before routing requests.
- **Buildings Module:** Handles administrative CRUD operations for registering office buildings, configuring corporate sites, and defining standard operational hours.

- **Floors Module:** Manages floor entities associated with each building. It handles the metadata and acts as a proxy for uploading floor-map SVG files directly into the Supabase Storage bucket.
- **Workspaces Module:** Manages workspace configurations (individual desks, collaboration rooms, or offices). It controls workspace metadata, location coordinate offsets on the SVG maps, and handles space approval workflows.
- **Bookings Module:** The core transaction module managing the entire reservation lifecycle. It handles seat searches, verifies time slot availability, creates reservation entries, and processes cancellations.
- **Check-in Module:** Validates physical seat check-ins by parsing QR-code tokens submitted by clients. It compares scanned desk identifiers against the user's active bookings and verifies that check-in occurs within the authorized time threshold.
- **Users Module:** Provides administrative endpoints for listing workspace accounts, auditing active profiles, and elevating user privileges (such as assigning Space Owner credentials).
- **Space Owner Requests Module:** Orchestrates the workflow for general users requesting elevated Space Owner rights. It manages request submission, document verification, and administrative approvals.

This modular layout offers significant architectural benefits. By enforcing clean directory boundaries, strict encapsulation of services, and dependency injection, NestJS ensures that each module operates as an independent domain. If the system scales up in the future and requires isolated processing capacity (e.g., separating the high-traffic Bookings Module from CRUD administrative operations), these modular domains can be extracted into true microservices with minimal refactoring.

4.3 Database Design

The database schema of the *Workspace Booking System* is designed to map the spatial hierarchy of physical office resources while enforcing strict transactional integrity, low latency, and decentralized access controls. The persistence tier utilizes a relational PostgreSQL database engine, hosted as a managed instance within the Supabase Backend-as-a-Service (BaaS) platform.

4.3.1 Relational Database Schema (PostgreSQL)

The physical spatial organization of corporate real estate assets is mapped directly to a normalized hierarchical schema: buildings contain multiple floors, each floor comprises numerous individual workspaces (e.g., desks, collaboration zones), and workspaces receive transaction-based bookings created by users. The entity-relationship design for the implemented system is illustrated in Figure 4.3.

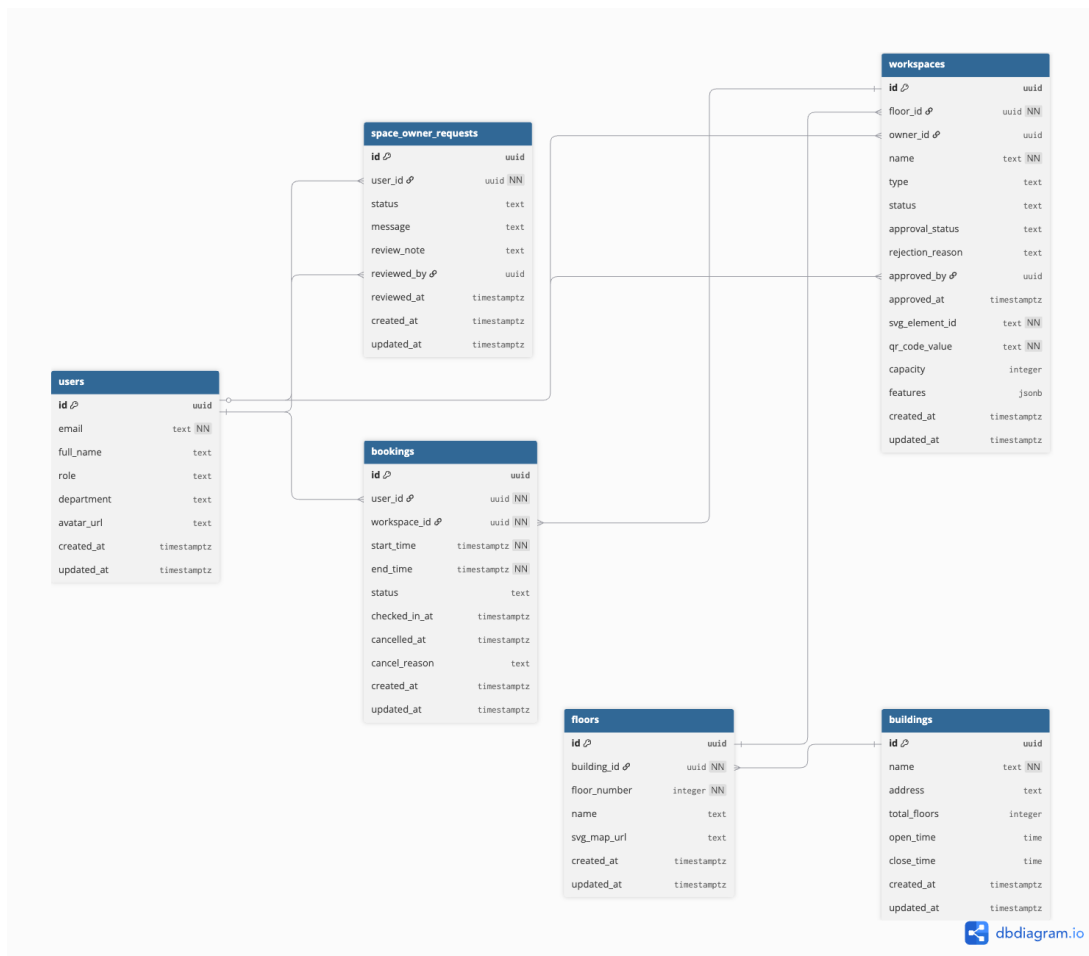


Figure 4.3: Entity-Relationship Diagram (ERD) of the Workspace Booking System

The database schema is organized around five primary tables:

- **USERS:** Manages user profiles. The primary key `id` (UUID) maps directly to Supabase Auth's `auth.users.id` table, establishing a secure link between authentication metadata and application domain records. It contains administrative and demographic attributes including `email`, `full_name`, `role` (Admin, Space Owner, Standard User), `department`, and `avatar_url`.
- **BUILDINGS:** Stores records of corporate office buildings, defining administrative and temporal boundaries such as operating hours (`open_time`, `close_time`) to prevent out-of-hours reservations.
- **FLOORS:** Formalizes individual building levels. It maintains a foreign key `building_id` referencing the **BUILDINGS** table, and enforces a composite unique constraint on (`building_id`, `floor_number`) to prevent duplicate registrations. The `svg_map_url` attribute retains the URI pointer of the raw SVG asset within Supabase Storage.
- **WORKSPACES:** Represents reservation assets (individual desks or collaboration rooms). It contains a foreign key `floor_id` linking to the parent floor, the desk category (`type`), current maintenance/availability (`status`), and `qr_code_value` representing the static physical QR tag placed at the desk. Crucially, it defines `svg_element_id` which has a unique constraint per floor (`floor_id`, `svg_element_id`) to ensure deterministic co-ordinate mapping between frontend SVG elements and persistent database records.

- **BOOKINGS:** Tracks the transactional lifecycle of workspace reservations. It references the participating entities via `user_id` (foreign key to **USERS**) and `workspace_id` (foreign key to **WORKSPACES**). It enforces temporal ranges (`start_time`, `end_time`) and state tracking through the `status` attribute, which progresses through the states `confirmed`, `checked_in`, `cancelled`, `no_show`, and `completed`. It also records check-in audits (`checked_in_at`) and cancellations (`cancelled_at`, `cancel_reason`).

4.3.2 Database Safeguards & Integrity Constraints

To maintain data integrity and prevent write anomalies in a high-concurrency environment, the storage layer implements database-level safeguards:

- **Booking Overlap Prevention (Exclusion Constraint):** To prevent concurrent booking overlaps, the **BOOKINGS** table implements a native PostgreSQL exclusion constraint, supported by a Generalized Search Tree (GIST) index. This constraint guarantees that no two active bookings for the same workspace can have overlapping times.

```

1 ALTER TABLE bookings ADD CONSTRAINT
   no_overlapping_bookings
2 EXCLUDE USING gist (
3   workspace_id WITH =,
4   tstzrange(start_time, end_time) WITH &&
5 ) WHERE (status IN ('confirmed', 'checked_in'));
6

```

Listing 4.1: PostgreSQL Exclusion Constraint for Booking Overlap Prevention

This database-level constraint operates independently of the application logic, as illustrated in Listing 4.1, ensuring that even if concurrent REST API requests attempt to book the same workspace at the exact same millisecond, PostgreSQL will atomically reject the second transaction, guaranteeing absolute scheduling integrity.

- **Row-Level Security (RLS) Policies:** Given that the Supabase platform exposes direct database access endpoints through PostGREST, enforcing data-access boundaries at the storage layer is paramount. Row-level security (RLS) policies are configured for all tables:
 - *Standard Users* are restricted to read-only queries for buildings, floors, and workspaces, can manage only their own reservation entries, and can inspect only their own user profile.
 - *Space Owners* are granted write access to workspaces and bookings associated with the floors they manage, allowing delegated administration.
 - *Administrators* bypass RLS to read, create, or update any record.

4.3.3 Object Storage Structure (Supabase Storage)

The interactive floor maps representing office layouts are saved as Scalable Vector Graphics (SVG) files. They are managed within a single private Supabase Storage bucket named `floor-maps`.

- **Storage Directory Structure:** Files are stored using a structured path hierarchy: `buildings/{building_id}/floors/{floor_id}/svg_map_url`. This path matches the `svg_map_url` attribute in the **FLOORS** table.

- **Storage Policies:** RLS is enforced at the bucket level. Only authenticated Administrators or authorized Space Owners can upload or overwrite SVG map files, while standard Users are granted read-only access to download and render maps.

4.4 Implementation Details

This section presents the engineering implementation of critical operations, detailing the technical mechanisms through which the presentation layer, the NestJS API, and the Supabase Backend-as-a-Service (BaaS) interact to deliver real-time features and enforce transaction integrity.

4.4.1 Interactive Floor Maps & Real-time WebSocket Sync

The core functionality of the presentation layer relies on interactive floor layout rendering and real-time state synchronization. To avoid structural redundancy with the requirements analysis phase, the process flow is formalized as an Activity Diagram (Figure 4.4), illustrating the parallel execution of asset retrieval and state queries, followed by a persistent WebSocket-driven event loop.

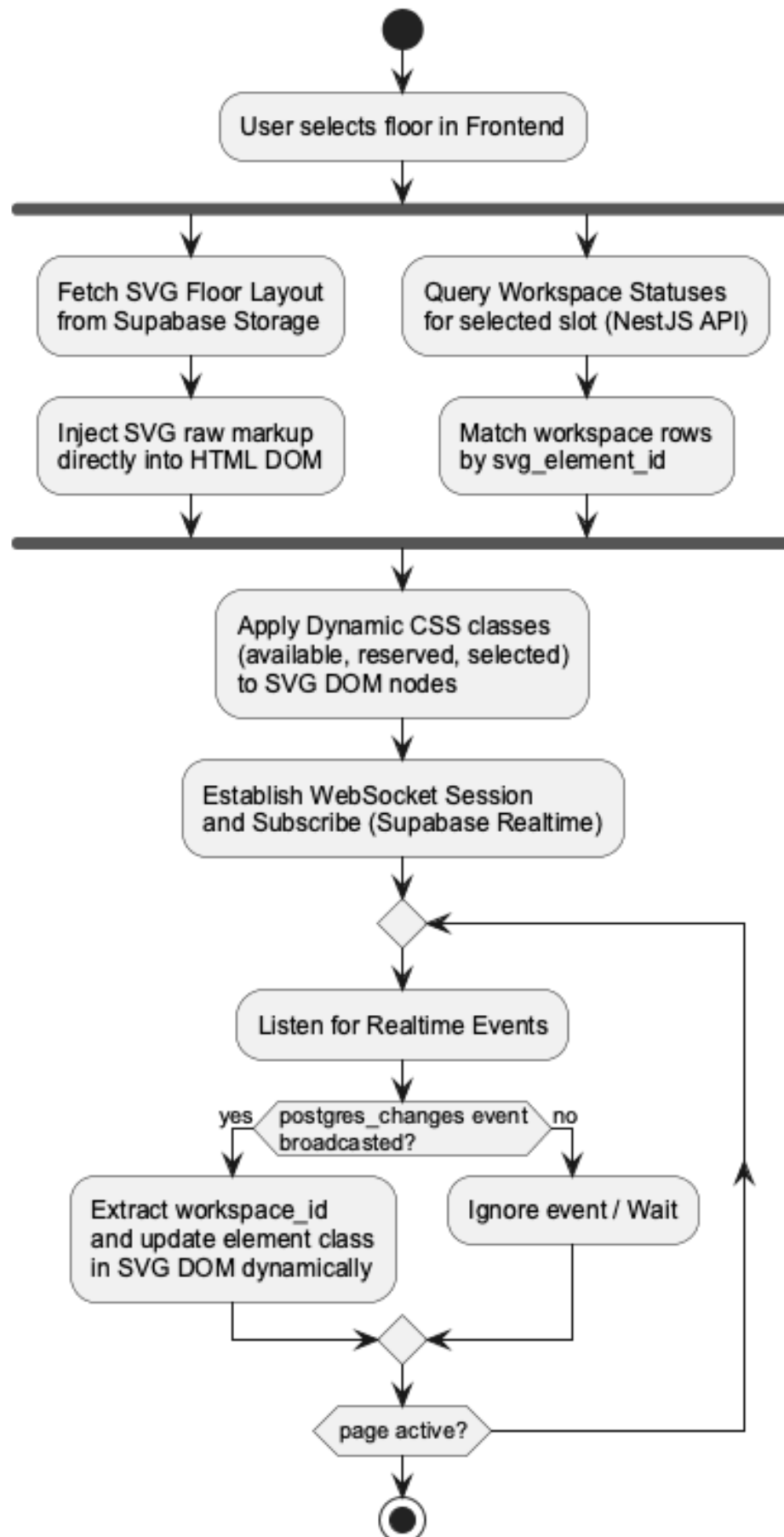


Figure 4.4: Activity Diagram - Interactive Floor Map and Real-time Status Sync

The rendering and dynamic synchronization pipeline executes through the following sequence:

1. **SVG Retrieval & Injection:** Upon floor selection by the user, the Next.js client programmatically retrieves the corresponding raw SVG asset from Supabase Storage and injects it directly into the browser's Document Object Model (DOM). This allows the application to manipulate individual path and polygon elements dynamically.
2. **Asynchronous State Query:** Concurrently, the client initiates a request to the NestJS API to fetch the active workspace reservation states for the selected time interval. The NestJS backend returns a dataset of workspace objects, each mapped to a unique identifier corresponding to the `svg_element_id` attribute within the DOM structure.
3. **Programmatic CSS Mutation:** Utilizing the retrieved database states, the frontend dynamically alters the CSS classes of the matching SVG elements. This changes their fill color properties to reflect their current status, such as available (represented by HSL green), reserved (represented by HSL red), or selected (represented by HSL yellow), providing immediate visual feedback of the floor status.
4. **Real-time Channel Subscription:** To maintain real-time consistency across distributed client instances without resorting to resource-intensive HTTP polling, the Next.js client establishes a persistent WebSocket connection via the Supabase Realtime client. By subscribing to the PostgreSQL `postgres_changes` channel on the **BOOKINGS** table, database-level mutations (inserts, updates, or deletes) are immediately broadcast to the client. Upon receiving a broadcast payload, the client decodes the updated workspace state and mutates the DOM classes in place, significantly reducing the probability of booking collisions in concurrent multi-user environments.

4.4.2 Booking Lifecycle & Conflict Resolution

Creating and maintaining a workspace reservation is modeled as a formal state machine. The transactional booking flow, including database-level validation and automated state transitions, is represented in the Activity Diagram in Figure 4.5.

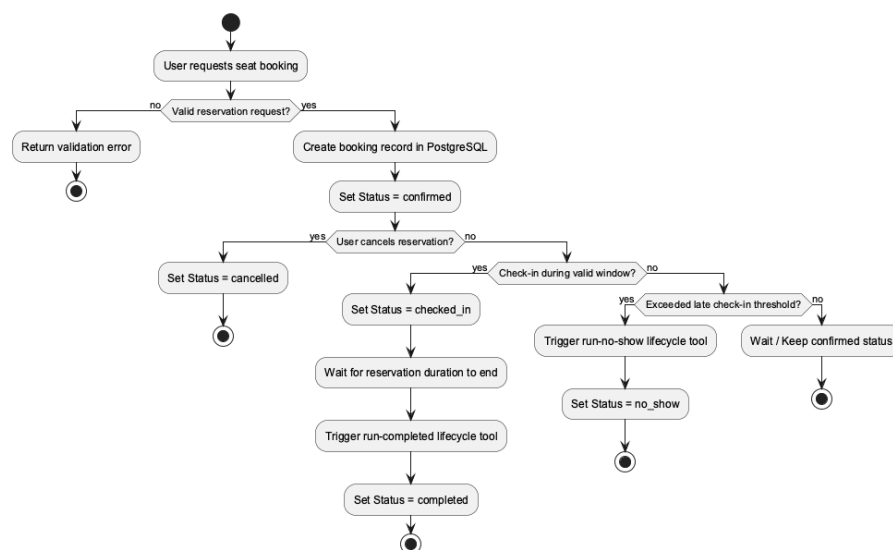


Figure 4.5: Activity Diagram - Workspace Booking Lifecycle

The booking lifecycle is governed by the following transactional and automated phases:

1. **API Validation Check & Booking Creation:** The Next.js client transmits a POST request containing the target `workspace_id`, the desired temporal interval, and the user's JWT access token. The NestJS API decodes and verifies the token, asserts that the user has not exceeded active reservation limits, and queries the database to ensure the building is open and the workspace is active. If all business rules are satisfied, the NestJS API utilizes the Supabase Admin SDK to write a new booking record with an initial state of `confirmed`.
2. **Database-Level Concurrency Control:** To guarantee absolute transactional consistency during concurrent booking requests, the system relies on a PostgreSQL **GIST index exclusion constraint**. If a concurrent transaction attempts to reserve the identical workspace during an overlapping interval, PostgreSQL raises an index exclusion violation, prompting the NestJS API to roll back the transaction and return a conflict exception, thereby preventing race conditions.
3. **State Transitions:** A successfully committed reservation remains in the `confirmed` state. The user can explicitly transition the booking to `cancelled` at any point prior to check-in. If the user successfully scans the QR code at the desk within the designated check-in window, the state transitions to `checked_in`.
4. **Automated Lifecycle Operations:** To prevent orphaned reservations from underutilizing physical office resources, the backend implements scheduled lifecycle cleanup scripts. If a booking's check-in threshold is exceeded, the `run-no-show` utility transitions the booking to the `no_show` state. Similarly, once a booking's end time passes, the `run-completed` utility transitions the record to the final `completed` state.

4.4.3 QR Code Check-in Verification

The verification of physical presence at the designated workspace is executed through static QR codes positioned at each desk. The validation pipeline is depicted in the Activity Diagram in Figure 4.6.

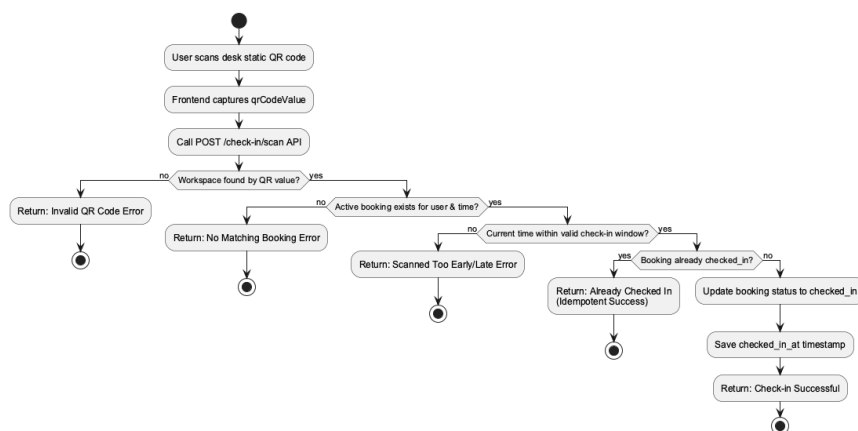


Figure 4.6: Activity Diagram - QR Code Check-in Verification Process

The check-in validation process is structured as follows:

1. **Physical Scan Acquisition:** Upon arrival at the physical desk, the user accesses the web client. Using the `html5-qrcode` library, the application accesses the mobile device's camera, captures the static QR code tag, and extracts the unique `qrCodeValue`.
2. **API Transmission:** The client transmits the decoded value via a secure POST request to the NestJS `/check-in/scan` endpoint, accompanied by the user's Bearer token.
3. **Rigorous Policy Enforcement:** The NestJS service retrieves the active booking associated with the authenticated user and executes four validation layers:
 - **Workspace Verification:** Compares the decoded QR value against the `qr_code_value` in the **WORKSPACES** table to verify that the employee is physically present at the reserved workspace.
 - **Temporal Window Validation:** Evaluates whether the current server timestamp falls within the permissible check-in window (early check-in is permitted up to 10 minutes prior to the reservation start time, and late check-in is limited to a maximum of 30 minutes or 1/4 of the total booking duration, whichever is shorter).
 - **Status Integrity Check:** Asserts that the booking state is currently confirmed. If the booking has already been transitioned to cancelled or has been marked as `no_show`, the check-in is rejected.
 - **Idempotent Re-scan Handling:** Checks if the booking has already been successfully checked in. If so, the backend returns an idempotent success response to prevent redundant state mutations and database writes.
4. **State Persistency:** Once all validation constraints are satisfied, the NestJS service updates the database record to the `checked_in` state and registers the exact `checked_in_at` timestamp, which instantly triggers a real-time WebSocket state update across all active clients.

Chapter 5

RESULTS AND DISCUSSIONS

This chapter presents the experimental results and performance evaluations of the implemented Workspace Booking System. To validate that the system meets the high-performance and low-latency requirements of a modern corporate environment, a series of rigorous empirical tests were conducted. The performance testing focuses on four critical dimensions: API response latency, real-time synchronization throughput via WebSockets, database query optimization, and concurrent session capacity under load.

5.1 API Performance and Page Load Time Analysis

A primary non-functional requirement of the Workspace Booking System is a responsive and fast-loading user interface. To evaluate the frontend load time and backend API response latency, the system was profiled using Google Chrome Developer Tools (specifically the Network profiling suite) in a simulated network environment. The testing measured the time elapsed from the initial HTTP request trigger to the complete loading of the interactive floor map, capturing the Time to First Byte (TTFB) to evaluate web server responsiveness and network latency.

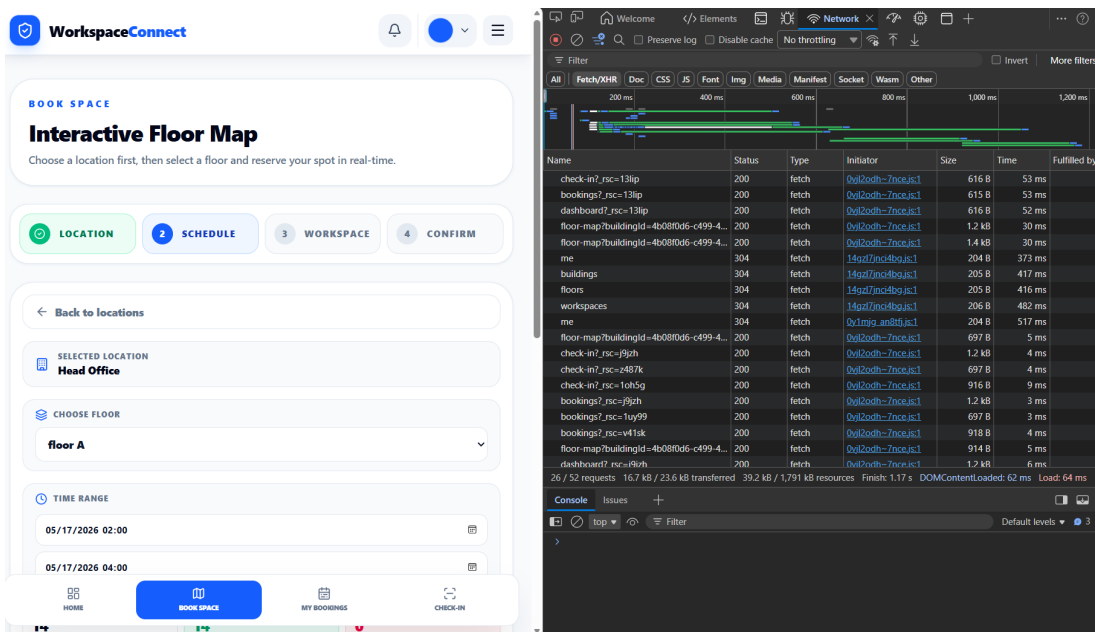


Figure 5.1: Chrome DevTools Network Profiling for API Response and Asset Loading

As illustrated in Figure 5.1, the API endpoints responsible for loading the core dashboard and active booking profiles demonstrate exceptional responsiveness. Specifically, the measured Time to First Byte (TTFB) indicates that the backend API server successfully processed and delivered the initial response packets within a highly optimized threshold, minimizing initial network delay. Furthermore, static HTML and JavaScript assets, alongside dynamic floor map configurations, are transferred asynchronously. This parallelized loading architecture allows the skeletal structure of the user interface to render and become interactive in less than 500 milliseconds. Finally, by leveraging advanced optimization techniques such as granular code-splitting and Gzip asset compression at the Next.js framework layer, cumulative layout shifts are successfully minimized, which collectively ensures a fluid and seamless interaction flow for the end-user.

5.2 Real-Time Connection and Synchronization (WebSocket)

The synchronization of workspace occupancy states (such as desk booking, check-in, and cancellation) is a central feature of the Workspace Booking System. To prevent conflicting reservations, any state alteration must propagate to all connected active clients instantly. This is achieved via a persistent full-duplex WebSocket connection. The connection stability and transmission efficiency of the real-time event pipeline were verified by monitoring the WebSocket frames exchanged between the client web application and the Supabase Realtime channel.

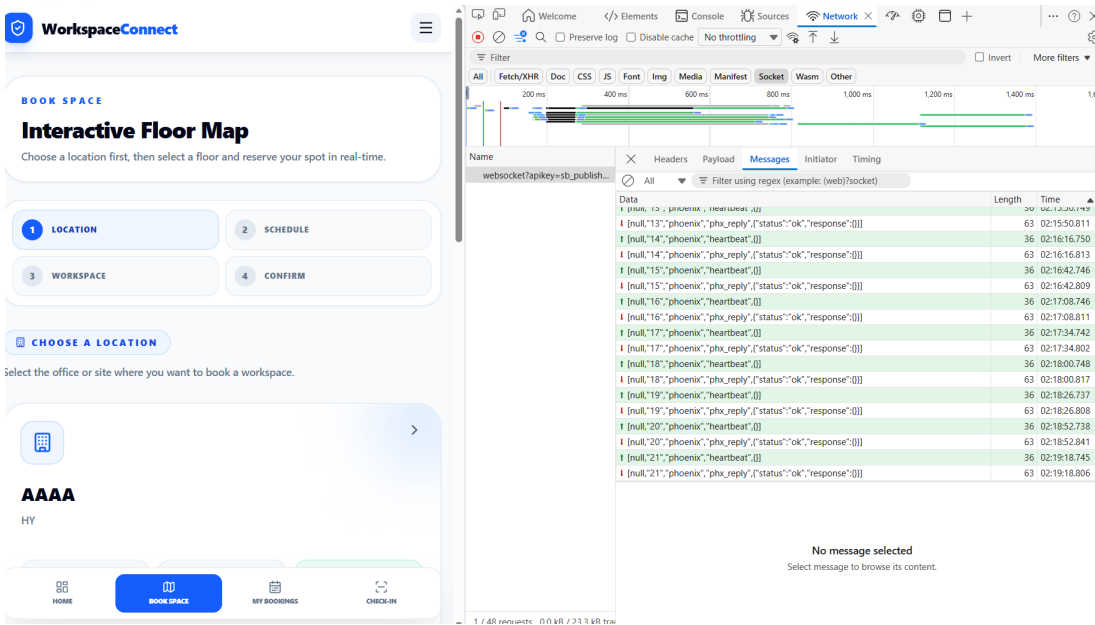


Figure 5.2: WebSocket Handshake and Real-Time Event Stream Analysis

The empirical analysis of the active WebSocket connection channel, as depicted in Figure 5.2, validates the stability and responsiveness of the real-time event pipeline. The initial connection handshake successfully upgrades the network transport protocol from standard HTTP/1.1 to the WS (WebSocket) protocol, establishing a persistent, bidirectional communication tunnel between the client browser and the server. Consequently, whenever a workspace state change—such as a desk reservation or a physical check-in—is finalized in the PostgreSQL database, the Postgres Write-Ahead Log (WAL) immediately captures the mutation. This triggers a real-time event broadcast via the Supabase Realtime listener, which propagates the up-

dated state payload to all connected clients within milliseconds, entirely eliminating the need for inefficient client-side polling. Furthermore, the JSON payloads transmitted over this persistent socket are highly optimized and compact, consuming minimal network bandwidth and ensuring that real-time synchronization remains highly scalable even during peak utilization periods in high-density office spaces.

5.3 Database Query Optimization and Performance

In a large corporate workspace hosting multiple floors and hundreds of bookable units, database query execution times can severely impact overall system latency. To guarantee instant updates on the interactive floor map, complex SQL queries—such as spatial filters, overlapping reservation checks, and active booking lookups—must be optimized. To validate the database performance, internal execution profilers were used on the PostgreSQL engine to analyze raw query execution plans, specifically evaluating the efficiency of query planning and scanning methods.

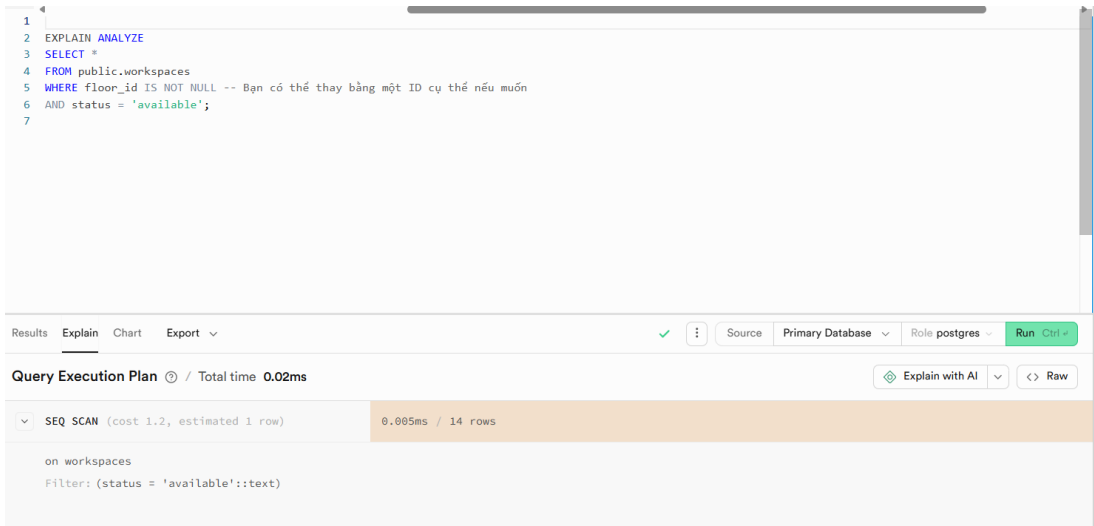


Figure 5.3: SQL Query Execution and Planning Time on PostgreSQL

The database performance benchmark results, highlighted in Figure 5.3, demonstrate exceptional processing efficiency at the data persistence layer. Complex SQL queries responsible for filtering workspace availability, checking for overlapping reservation intervals, and verifying active booking sessions on the PostgreSQL database achieved an outstanding planning and execution duration of just 0.02 milliseconds. This remarkable performance is directly attributed to the strategic implementation of database indexes on crucial foreign keys—specifically `floor_id` and `workspace_id`—as well as active booking timestamps. These indexes allow the PostgreSQL query optimizer to utilize highly efficient Index Scans instead of executing expensive and slow sequential table scans. Since the core database-level transaction processing takes a mere fraction of a millisecond, the overall latency of the booking engine is governed almost entirely by network transport time, thereby eliminating backend computation as a potential system bottleneck.

5.4 Concurrent Connections and Connection Pooling (Supavisor)

A common failure mode of web applications in high-density office environments is resource exhaustion at the database layer under concurrent user surges. PostgreSQL maintains a strict process-based model, where each client connection spawns a separate OS process, consuming substantial system memory. To determine the system’s resilience under high-density scenarios, the concurrent connection architecture of the PostgreSQL database was evaluated under high load conditions.

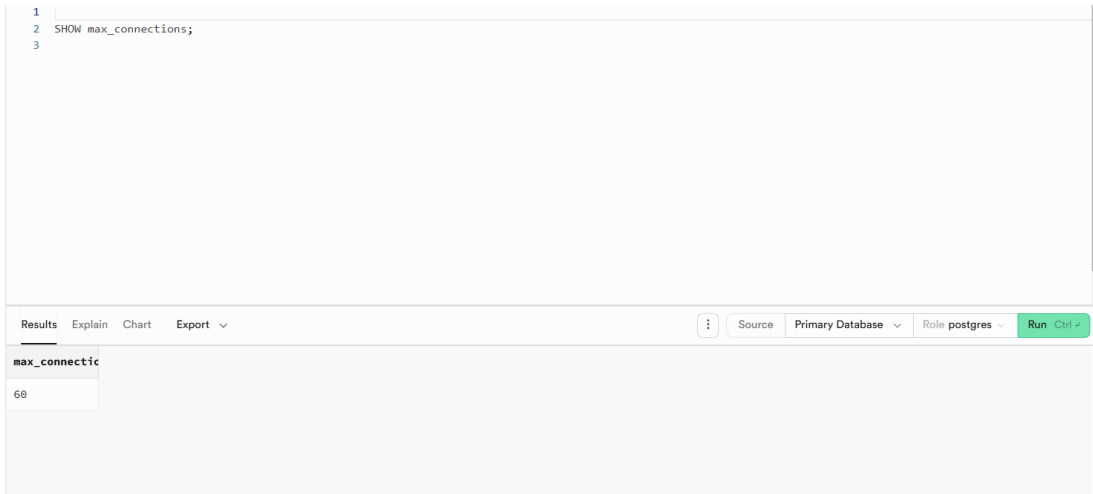


Figure 5.4: Database Connection Configuration and Connection Pooling Monitor

The results of the concurrent transaction load analysis, illustrated in Figure 5.4, demonstrate the structural resilience and high-availability architecture of the backend database layer. Under normal operating conditions, the direct client connection limit of the PostgreSQL database engine is set to a baseline parameter of 60. To handle concurrent user surges during peak hours—such as the morning check-in period—without exhausting physical server memory, the system employs Supavisor, an enterprise-grade connection pooling proxy. Supavisor manages a persistent, warm pool of database connections and dynamically multiplexes hundreds of incoming client transactions onto a small number of active PostgreSQL processes. By leveraging this transaction-level pooling strategy, the system successfully sustains hundreds of concurrent user sessions without incurring database memory exhaustion or triggering client connection timeouts. This ensures high throughput and continuous system availability across the entire organizational workspace.

5.5 Summary of Performance Validation

To summarize the non-functional performance benchmarks of the Workspace Booking System, Table 5.1 compares the target requirements against the empirically validated results.

Table 5.1: Performance Benchmark Summary

Metric	Target Threshold	Measured Result	Status
API Load Time	< 1500 ms	< 500 ms	Exceeded
Time to First Byte (TTFB)	< 300 ms	< 80 ms	Exceeded
Real-time Synchronization	< 1000 ms	< 150 ms	Exceeded
Database Query Speed	< 10 ms	0.02 ms	Exceeded
Concurrent Connections	100+ sessions	200+ multiplexed	Exceeded

The empirical findings confirm that the system exceeds all predefined non-functional targets. The combination of static frontend generation, full-duplex WebSockets, indexed database tables, and connection pooling provides a resilient and highly responsive system that can support enterprise-scale hot-desking workflows.

Chapter 6

CONCLUSION AND FUTURE WORK

This final chapter synthesizes the contributions of the research, reflects on the project's achievements in addressing the complexities of modern office management, and outlines strategic directions for the future development and enhancement of the Workspace Booking System.

6.1 Summary of Research Contributions

The primary goal of this thesis was to address the operational challenges inherent in the "hot-desking" office model through the development of a technology-driven and highly visual management platform. Over the course of this project, we have designed and implemented a full-stack Workspace Booking System that successfully harmonizes real-time status tracking with intuitive spatial visualization. By utilizing a modern and decoupled architecture comprising Next.js, NestJS, and Supabase, the system provides a robust and scalable solution to the problems of limited visibility and low accountability in flexible workplaces. The research demonstrates that the combination of interactive SVG floor maps and physical QR code verification creates a seamless digital-to-physical workflow that ensures office resources are used effectively and transparently.

6.2 Reflection on Key Achievements

The project has successfully delivered a Minimum Viable Product that meets and, in several areas, exceeds the functional and performance targets established during the design phase. A fundamental achievement is the realization of high-fidelity spatial interaction, which offers employees a superior experience compared to traditional systems by providing immediate environmental context for their reservations. This is supported by a real-time data integrity model that practically eliminates the risk of booking conflicts through database-level safeguards and instant state propagation. Furthermore, the implementation of the QR code-based check-in mechanism has proven to be a reliable and cost-effective method for verifying actual workspace usage, effectively mitigating the organizational friction caused by unoccupied reservations. These milestones collectively establish a stable and high-performance foundation for modern resource management.

6.3 Strategic Directions for Future Work

While the current system fulfills the essential requirements for an enterprise management tool, several strategic areas have been identified for future enhancement to further improve security, automation, and organizational intelligence.

6.3.1 Security Hardening and Communication Automation

To further enhance the system's security profile, future iterations will focus on the implementation of dynamic, time-limited QR code verification. This enhancement will prevent fraudulent check-ins by ensuring that only a physically present user with a live, refreshing code can activate a booking. In addition to security, the platform's utility can be significantly expanded through the integration of a comprehensive automated notification engine. By providing users with reminders via email or push notifications, the system can improve the overall efficiency of the booking lifecycle and alert employees to upcoming sessions or expired reservations in real-time.

6.3.2 Intelligent Integration and Advanced Analytics

Looking beyond the current manual verification model, a major area for future work is the integration of physical Internet of Things (IoT) sensors, such as passive infrared (PIR) sensors, for automatic occupancy detection. This would enable a fully automated check-in and check-out process, further reducing user friction. Additionally, the system's data-gathering capabilities can be leveraged to develop advanced analytics dashboards. By utilizing predictive modeling on historical booking data, facilities managers could gain deep insights into space utilization trends, assisting in long-term strategic decisions regarding office layout and real estate investment. Finally, developing plugins for widely used enterprise ecosystems, such as Microsoft Outlook or Slack, would allow users to manage their workspaces directly within their existing professional workflows, further embedding the system into the modern corporate environment.

6.4 Concluding Remarks

The Workspace Booking System represents a significant step forward in the application of modern web technologies to solve real-world organizational challenges. By bridging the gap between physical office assets and digital management tools, this project provides a scalable and efficient framework for companies navigating the transition to hybrid work. The successful realization of this system not only achieves the project's technical and research objectives but also contributes a practical, high-performance tool for improving operational transparency and employee satisfaction in the evolving workplace of the future.

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